

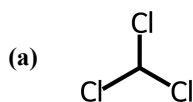
OPTICAL ISOMERISM — SOLUTIONS

Part A — Elements of symmetry

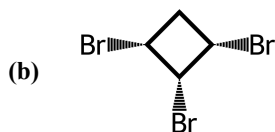
1. Answer: 24

For each of the following molecules mark the elements of symmetry present among POS (plane of symmetry), COS (centre of symmetry) and AOS (axis of symmetry, a proper rotation axis C_n with $n \geq 2$). Find the total number of elements marked over all 14 molecules.

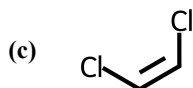
Method: list the symmetry elements of every structure. A plane of symmetry (POS) divides the molecule into two mirror-image halves; a centre of symmetry (COS) maps every atom to an identical atom through the centre; an axis of symmetry (AOS) is a proper rotation axis C_n ($n \geq 2$) after which the molecule looks unchanged; an alternate axis (AAOS) is a rotation-reflection axis S_n ($n \geq 3$). Rings are taken as planar with wedge groups above and dashed groups below the ring plane; for each structure decide which of the four elements exist, then add up the marks.



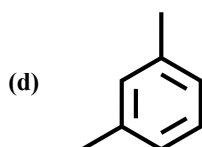
chloroform: POS=1, COS=0, AOS=1 → 2



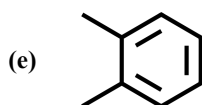
4-membered ring: Br on C2 (below the ring), Br on C3 (below the ring), Br on C4 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral. Marks: POS=1, COS=0, AOS=0 → 1



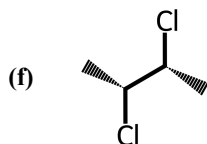
(Z)-1,2-dichloroethene: POS=1, COS=0, AOS=1 → 2



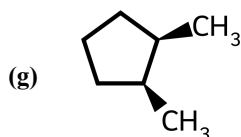
m-xylene: POS=1, COS=0, AOS=1 → 2



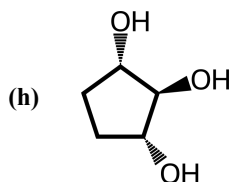
o-xylene: POS=1, COS=0, AOS=1 → 2



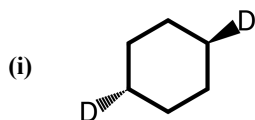
(R,R)/(S,S)-2,3-dichlorobutane (time-averaged): POS=0, COS=0, AOS=1 → 1



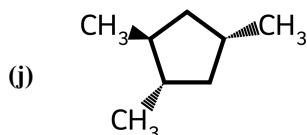
5-membered ring: CH₃ on C4 (above the ring), CH₃ on C5 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral. Marks: POS=1, COS=0, AOS=0 → 1



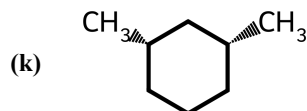
5-membered ring: OH on C1 (below the ring), OH on C4 (below the ring), OH on C5 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral. Marks: POS=1, COS=0, AOS=0 → 1



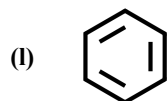
6-membered ring: D on C3 (below the ring), D on C6 (above the ring). Found: 1 mirror plane(s); an inversion centre (opposite carbons carry the same group on opposite faces); 1 proper axis/axes; no alternate axis; so it is achiral. Marks: POS=1, COS=1, AOS=1 → 3



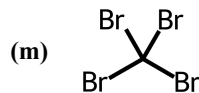
5-membered ring: CH₃ on C2 (above the ring), CH₃ on C3 (below the ring), CH₃ on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral. Marks: POS=0, COS=0, AOS=0 → 0



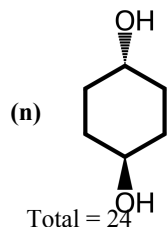
6-membered ring: CH₃ on C2 (below the ring), CH₃ on C6 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral. Marks: POS=1, COS=0, AOS=0 → 1



benzene: POS=1, COS=1, AOS=1 → 3



carbon tetrabromide: POS=1, COS=0, AOS=1 → 2

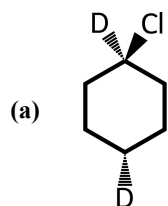


6-membered ring: OH on C1 (below the ring), OH on C4 (above the ring). Found: 1 mirror plane(s); an inversion centre (opposite carbons carry the same group on opposite faces); 1 proper axis/axes; no alternate axis; so it is achiral. Marks: POS=1, COS=1, AOS=1 → 3

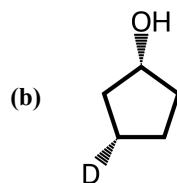
2. Answer: 3

How many of the following are optically active (chiral)?

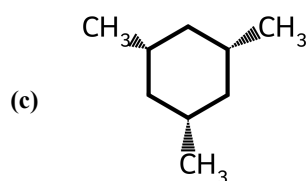
Method: a molecule is chiral only when it has no improper symmetry element (plane, centre, S_n). Find the symmetry elements of each structure; if none of POS, COS and AAOS is present the structure is chiral.



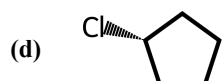
6-membered ring: Cl on C1 (above the ring), D on C1 (below the ring), D on C4 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



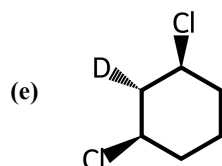
5-membered ring: OH on C1 (below the ring), D on C3 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → counts



6-membered ring: CH₃ on C2 (below the ring), CH₃ on C4 (below the ring), CH₃ on C6 (below the ring). Found: 3 mirror plane(s); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is achiral → does not count



5-membered ring: Cl on C2 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



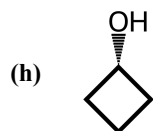
6-membered ring: Cl on C1 (above the ring), D on C2 (below the ring), Cl on C3 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



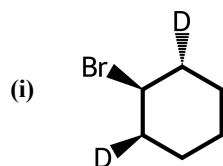
4-membered ring: Cl on C2 (above the ring), CH₃ on C4 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



5-membered ring: Cl on C5 (above the ring), D on C5 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



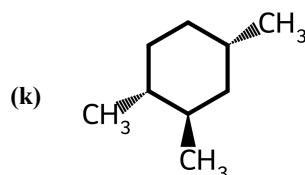
4-membered ring: OH on C1 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



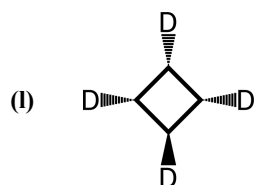
6-membered ring: D on C1 (below the ring), Br on C2 (above the ring), D on C3 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → counts



4-membered ring: OH on C4 (above the ring), OH on C4 (below the ring). Found: 2 mirror plane(s); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is achiral → does not count



6-membered ring: CH3 on C3 (below the ring), CH3 on C4 (above the ring), CH3 on C6 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → counts



4-membered ring: D on C1 (below the ring), D on C2 (below the ring), D on C3 (above the ring), D on C4 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count

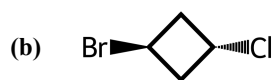
3. Answer: 5

Find the total number of planes of symmetry present in the compounds (a)–(f) taken together.

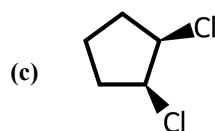
Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.



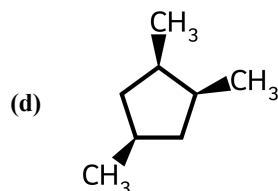
4-membered ring: Br on C4 (above the ring), CH3 on C4 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → 1



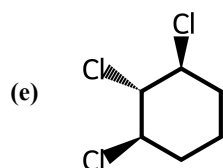
4-membered ring: Br on C2 (above the ring), Cl on C4 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → 1



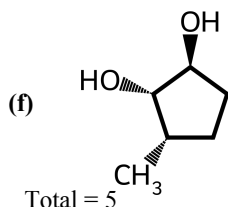
5-membered ring: Cl on C4 (above the ring), Cl on C5 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → 1



5-membered ring: CH3 on C1 (above the ring), CH3 on C3 (above the ring), CH3 on C5 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → 1



6-membered ring: Cl on C1 (above the ring), Cl on C2 (below the ring), Cl on C3 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → 1



5-membered ring: OH on C1 (above the ring), OH on C2 (below the ring), CH₃ on C3 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → 0

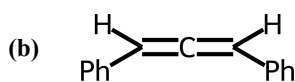
4. Answer: 8

How many of the following possess an ALTERNATE axis of symmetry (a rotation–reflection axis S_n with $n \geq 3$)?

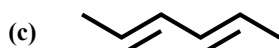
Method: an S_n axis is a rotation by $360^\circ/n$ followed by reflection in the plane perpendicular to the axis (rotation–reflection axis). S_1 is a plane and S_2 a centre, so only $n \geq 3$ counts here. Alternate axes occur in these point groups: D_{2d} (dihedral with diagonal planes: S_4), D_{3d} (S_6), D_{4d} (S_8), D_{5d} (S_{10}), D_{4h} (S_4), D_{5h} (S_5), D_{6h} (S_6 and S_3), C_{3h} (S_3), C_{4h} (S_4), C_{6h} (S_6 and S_3), S_4 (S_4), S_6 (S_6), T_d (tetrahedral: S_4) and O_h (octahedral: S_4 and S_6). The point groups C_1 , C_s , C_i , C_n , C_{nv} , D_n , C_{2h} and D_{2h} have no S_n with $n \geq 3$. Identify the point group of each structure and list its alternate axes.



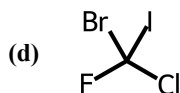
norbornane (C_{2v}): no



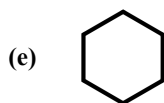
1,3-diphenylallene (C_2): no



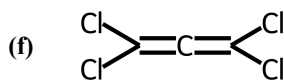
(E,E)-hexa-2,4-diene (planar, C_{2h}): no



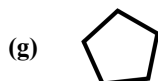
bromochlorofluoroiodomethane (C_1): no



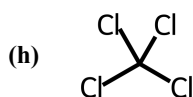
cyclohexane (planar, D_{6h}): yes



1,1,3,3-tetrachloroallene (D_{2d}): yes



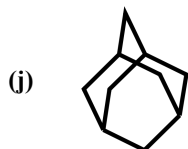
cyclopentane (planar, D_{5h}): yes



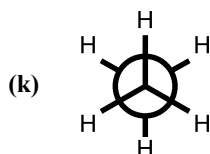
tetrachloromethane (T_d): yes



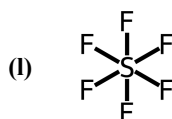
spiro[3.3]heptane (D_{2d}): yes



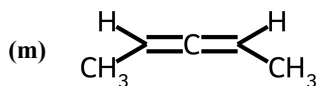
adamantane (T_d): yes



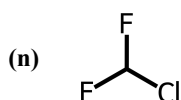
staggered ethane (D_{3d}): yes



sulfur hexafluoride (O_h): yes



penta-2,3-diene (C₂): no



chlorodifluoromethane (C_s): no

5. Answer: 10

Read the following 14 statements on symmetry elements and find how many of them are correct.

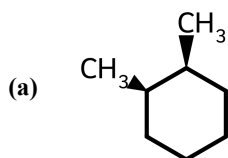
Method: judge every statement independently using the definitions of chirality, symmetry elements, configuration and properties of stereoisomers; the explanation given with each statement states why it is true or false.

- (1) True — A plane is S₁, an improper axis.
- (2) False — A centre is S₂; any S_n axis makes the molecule achiral.
- (3) True — Only proper axes: no improper element.
- (4) True — Definition of the rotation–reflection axis.
- (5) True — Any S_n axis excludes chirality.
- (6) True — T_d point group.
- (7) True — D_{6h}: $\sigma_h + 3\sigma_v + 3\sigma_d = 7$ planes.
- (8) True — D_{6h}.
- (9) True — Plane through C₂ and C₅.
- (10) False — It has only a C₂ axis and is chiral.
- (11) True — Inversion maps C₁ to C₄ and up to down.
- (12) False — Both methyls are on the same face, so inversion does not map them onto each other.
- (13) True — D_{3d}.
- (14) False — D_{2d}: planes, C₂ axes and an S₄ axis, but no centre.

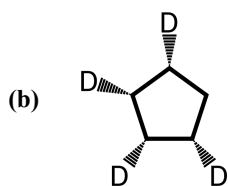
6. Answer: 5

How many of the following are ASYMMETRIC (possess no element of symmetry at all)?

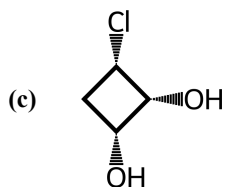
Method: an asymmetric molecule has no symmetry element at all except the identity (point group C₁). Test for planes, a centre, proper and alternate axes; count only those with none.



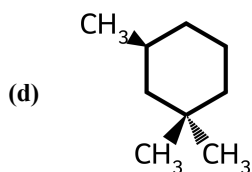
6-membered ring: CH₃ on C₁ (above the ring), CH₃ on C₂ (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



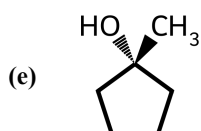
5-membered ring: D on C₁ (below the ring), D on C₂ (below the ring), D on C₃ (below the ring), D on C₄ (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



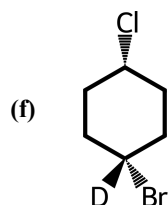
4-membered ring: Cl on C₁ (below the ring), OH on C₃ (below the ring), OH on C₄ (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → counts



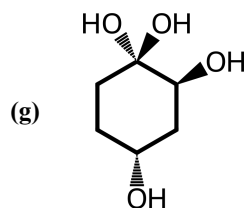
6-membered ring: CH₃ on C₂ (above the ring), CH₃ on C₄ (above the ring), CH₃ on C₄ (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → counts



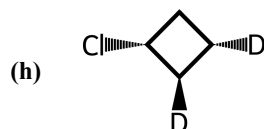
5-membered ring: CH₃ on C₁ (above the ring), OH on C₁ (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



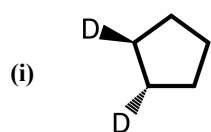
6-membered ring: Cl on C1 (below the ring), D on C4 (above the ring), Br on C4 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



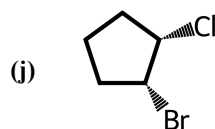
6-membered ring: OH on C1 (above the ring), OH on C1 (below the ring), OH on C2 (above the ring), OH on C4 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → counts



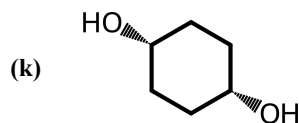
4-membered ring: Cl on C2 (below the ring), D on C3 (above the ring), D on C4 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → counts



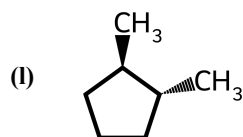
5-membered ring: D on C2 (above the ring), D on C3 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → does not count



5-membered ring: Br on C4 (below the ring), Cl on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → counts



6-membered ring: OH on C2 (below the ring), OH on C5 (below the ring). Found: 2 mirror plane(s); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is achiral → does not count

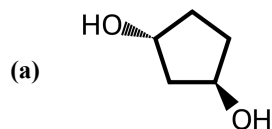


5-membered ring: CH₃ on C1 (above the ring), CH₃ on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → does not count

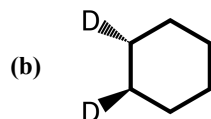
7. Answer: 4

How many of the following are DISSYMMETRIC but not asymmetric (chiral, yet possessing a proper axis of symmetry)?

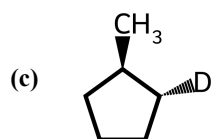
Method: a proper axis C_n rotates the molecule by $360^\circ/n$ into an identical orientation. Check for such rotations (in planar-ring drawings the ring normal and the in-plane axes through opposite atoms or bond mid-points must map every wedge/dash group onto an identical one) and count or add as asked.



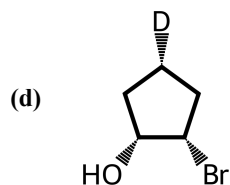
5-membered ring: OH on C2 (below the ring), OH on C4 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → counts



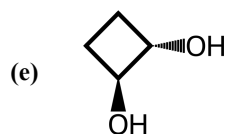
6-membered ring: D on C2 (below the ring), D on C3 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → counts



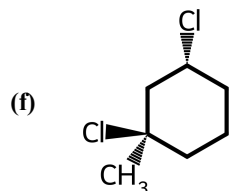
5-membered ring: CH₃ on C1 (above the ring), D on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



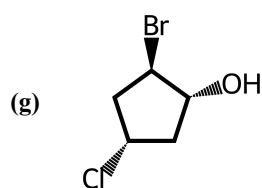
5-membered ring: D on C1 (below the ring), OH on C3 (below the ring), Br on C4 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



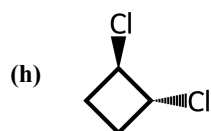
4-membered ring: OH on C3 (above the ring), OH on C4 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → counts



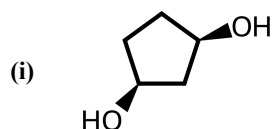
6-membered ring: Cl on C1 (below the ring), Cl on C3 (above the ring), CH3 on C3 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



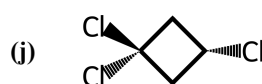
5-membered ring: Br on C1 (above the ring), Cl on C3 (below the ring), OH on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



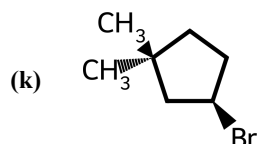
4-membered ring: Cl on C1 (above the ring), Cl on C4 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → counts



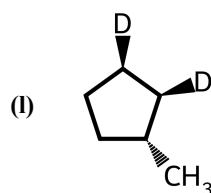
5-membered ring: OH on C3 (above the ring), OH on C5 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



4-membered ring: Cl on C2 (above the ring), Cl on C2 (below the ring), Cl on C4 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



5-membered ring: CH3 on C2 (above the ring), CH3 on C2 (below the ring), Br on C4 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count

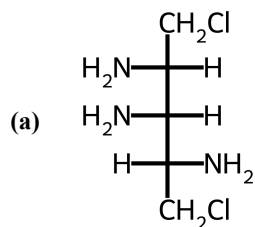


5-membered ring: D on C1 (above the ring), CH3 on C4 (below the ring), D on C5 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count

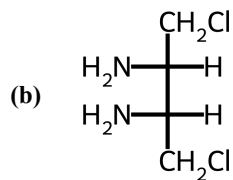
8. Answer: 4

How many of the following Fischer projections (eclipsed conformation, as drawn) possess a PLANE of symmetry?

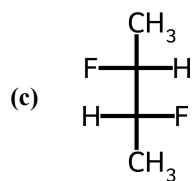
Method: look for a mirror plane that maps the molecule onto itself. In planar-ring drawings the ring plane itself is a plane only if no group is on one face; vertical planes must map every wedge group onto an identical wedge group and every dashed group onto an identical dashed group. Check each structure and count those that have one (or add the planes if the question asks for a total).



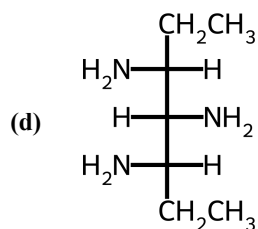
chiral: no plane $\rightarrow$ does not count



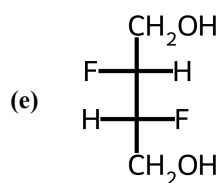
achiral (meso): plane of symmetry $\rightarrow$ counts



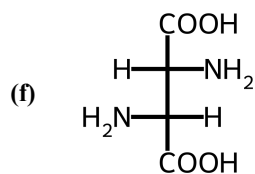
chiral: no plane; C2 axis $\rightarrow$ does not count



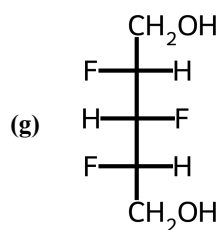
achiral (meso): plane of symmetry $\rightarrow$ counts



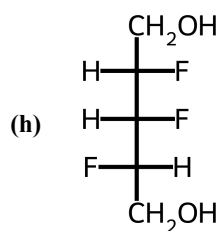
chiral: no plane; C2 axis $\rightarrow$ does not count



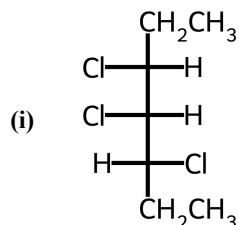
chiral: no plane; C2 axis $\rightarrow$ does not count



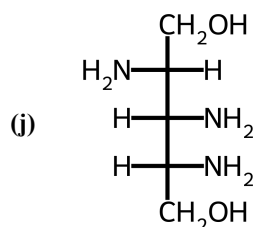
achiral (meso): plane of symmetry $\rightarrow$ counts



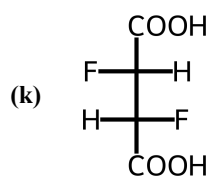
chiral: no plane $\rightarrow$ does not count



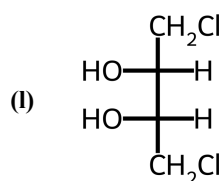
chiral: no plane → does not count



chiral: no plane → does not count



chiral: no plane; C₂ axis → does not count

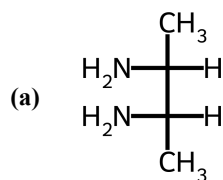


achiral (meso): plane of symmetry → counts

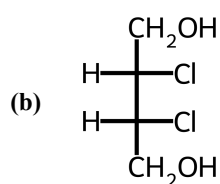
9. Answer: 4

How many of the following Fischer projections (eclipsed conformation, as drawn) possess a C₂ axis of symmetry perpendicular to the plane of the paper through the middle of the chain?

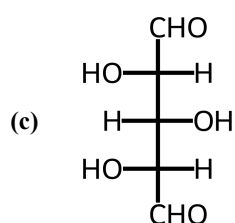
Method: a proper axis C_n rotates the molecule by 360°/n into an identical orientation. Check for such rotations (in planar-ring drawings the ring normal and the in-plane axes through opposite atoms or bond mid-points must map every wedge/dash group onto an identical one) and count or add as asked.



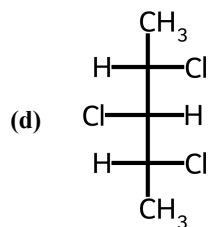
achiral (meso): plane of symmetry → does not count



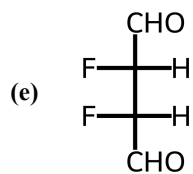
achiral (meso): plane of symmetry → does not count



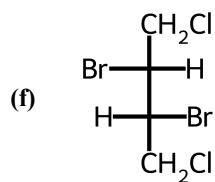
achiral (meso): plane of symmetry → does not count



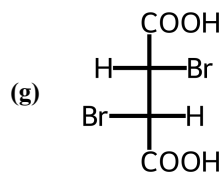
achiral (meso): plane of symmetry → does not count



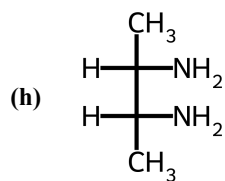
achiral (meso): plane of symmetry → does not count



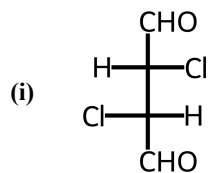
chiral: no plane; C₂ axis → counts



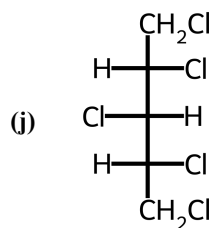
chiral: no plane; C₂ axis → counts



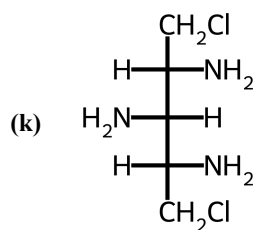
achiral (meso): plane of symmetry → does not count



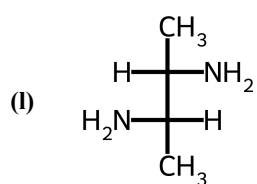
chiral: no plane; C₂ axis → counts



achiral (meso): plane of symmetry → does not count



achiral (meso): plane of symmetry → does not count



chiral: no plane; C₂ axis → counts

10. Answer: 16

Mark POS (plane), COS (centre), AOS (proper axis, C_n with $n \geq 2$) and AAOS (alternate axis, S_n with $n \geq 3$) for each of the following species. Find the total number of marks over all eight species.

Method: list the symmetry elements of every structure. A plane of symmetry (POS) divides the molecule into two mirror-image halves; a centre of symmetry (COS) maps every atom to an identical atom through the centre; an axis of symmetry (AOS) is a proper rotation axis C_n ($n \geq 2$) after which the molecule looks unchanged; an alternate axis (AAOS) is a rotation-reflection axis S_n ($n \geq 3$). Rings are taken as planar with wedge groups above and dashed groups below the ring plane; for each structure decide which of the four elements exist, then add up the marks.

- | | | |
|------------|--|---|
| (a) | envelope cyclopentane | envelope cyclopentane: POS=1, COS=0, AOS=0, AAOS=0 → 1 |
| (b) | anti-butane | anti-butane: POS=1, COS=1, AOS=1, AAOS=0 → 3 |
| (c) | trans-1,2-dimethylcyclohexane (diequatorial chair) | trans-1,2-dimethylcyclohexane (diequatorial chair): POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (d) | trans-decalin | trans-decalin: POS=1, COS=1, AOS=1, AAOS=0 → 3 |
| (e) | staggered ethane | staggered ethane: POS=1, COS=1, AOS=1, AAOS=1 → 4 |
| (f) | cis-1,2-dimethylcyclohexane (time-averaged) | cis-1,2-dimethylcyclohexane (time-averaged): POS=1, COS=0, AOS=0, AAOS=0 → 1 |
| (g) | gauche-1,2-dichloroethane | gauche-1,2-dichloroethane: POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (h) | boat cyclohexane | boat cyclohexane: POS=1, COS=0, AOS=1, AAOS=0 → 2 |
| Total = 16 | | |

11. Answer: 4

How many of the following species are chiral (no plane, no centre and no alternate axis of symmetry)?

Method: an S_n axis is a rotation by $360^\circ/n$ followed by reflection in the plane perpendicular to the axis (rotation-reflection axis). S_1 is a plane and S_2 a centre, so only $n \geq 3$ counts here. Alternate axes occur in these point groups: D_{2d} (dihedral with diagonal planes: S_4), D_{3d} (S_6), D_{4d} (S_8), D_{5d} (S_{10}), D_{4h} (S_4), D_{5h} (S_5), D_{6h} (S_6 and S_3), C_{3h} (S_3), C_{4h} (S_4), C_{6h} (S_6 and S_3), S_4 (S_4), S_6 (S_6), T_d (tetrahedral: S_4) and O_h (octahedral: S_4 and S_6). The point groups C_1 , C_s , C_i , C_n , C_{nv} , D_n , C_{2h} and D_{2h} have no S_n with $n \geq 3$. Identify the point group of each structure and list its alternate axes.

- | | | |
|-----|--|---|
| (a) | anti-1,2-dichloroethane | anti-1,2-dichloroethane: POS=1, COS=1, AOS=1, AAOS=0 → 0 |
| (b) | trans-1,4-dimethylcyclohexane (diequatorial chair) | trans-1,4-dimethylcyclohexane (diequatorial chair): POS=1, COS=1, AOS=1, AAOS=0 → 0 |
| (c) | (R,R)-tartaric acid (anti conformation) | (R,R)-tartaric acid (anti conformation): POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (d) | chair cyclohexane | chair cyclohexane: POS=1, COS=1, AOS=1, AAOS=1 → 0 |
| (e) | gauche-butane | gauche-butane: POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (f) | adamantane | adamantane: POS=1, COS=0, AOS=1, AAOS=1 → 0 |
| (g) | half-chair cyclohexene | half-chair cyclohexene: POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (h) | syn-eclipsed butane | syn-eclipsed butane: POS=1, COS=0, AOS=1, AAOS=0 → 0 |
| (i) | anti-butane | anti-butane: POS=1, COS=1, AOS=1, AAOS=0 → 0 |
| (j) | trans-decalin | trans-decalin: POS=1, COS=1, AOS=1, AAOS=0 → 0 |
| (k) | cyclooctatetraene (tub) | cyclooctatetraene (tub): POS=1, COS=0, AOS=1, AAOS=1 → 0 |
| (l) | trans-cyclooctene | trans-cyclooctene: POS=0, COS=0, AOS=1, AAOS=0 → 1 |

12. Answer: 6

How many of the following species possess a C₂ (or higher) axis of symmetry but NO plane of symmetry?

Method: look for a mirror plane that maps the molecule onto itself. In planar-ring drawings the ring plane itself is a plane only if no group is on one face; vertical planes must map every wedge group onto an identical wedge group and every dashed group onto an identical dashed group. Check each structure and count those that have one (or add the planes if the question asks for a total).

- | | | |
|-----|--|---|
| (a) | twistane | twistane: POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (b) | syn-eclipsed butane | syn-eclipsed butane: POS=1, COS=0, AOS=1, AAOS=0 → 0 |
| (c) | boat cyclohexane | boat cyclohexane: POS=1, COS=0, AOS=1, AAOS=0 → 0 |
| (d) | staggered ethane | staggered ethane: POS=1, COS=1, AOS=1, AAOS=1 → 0 |
| (e) | gauche-1,2-dichloroethane | gauche-1,2-dichloroethane: POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (f) | hexahelicene | hexahelicene: POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (g) | cis-cyclooctene (crown/boat-chair) | cis-cyclooctene (crown/boat-chair): POS=1, COS=0, AOS=0, AAOS=0 → 0 |
| (h) | chair cyclohexane | chair cyclohexane: POS=1, COS=1, AOS=1, AAOS=1 → 0 |
| (i) | twist-boat cyclohexane | twist-boat cyclohexane: POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (j) | cis-1,3-dimethylcyclohexane (diequatorial chair) | cis-1,3-dimethylcyclohexane (diequatorial chair): POS=1, COS=0, AOS=0, AAOS=0 → 0 |
| (k) | half-chair cyclohexene | half-chair cyclohexene: POS=0, COS=0, AOS=1, AAOS=0 → 1 |
| (l) | trans-cyclooctene | trans-cyclooctene: POS=0, COS=0, AOS=1, AAOS=0 → 1 |

13. Answer: 3

How many of the following species possess BOTH a plane and a centre of symmetry?

Method: a centre of symmetry (inversion centre $i = S_2$) must send every atom through the centre to an identical atom. In ring compounds this means a group on one face must be matched by the same group on the opposite face of the diametrically opposite carbon, so only rings with an even number of members can have it.

- | | | |
|-----|--|---|
| (a) | trans-1,4-dimethylcyclohexane (diequatorial chair) | trans-1,4-dimethylcyclohexane (diequatorial chair): POS=1, COS=1, AOS=1, AAOS=0 → 1 |
| (b) | staggered ethane | staggered ethane: POS=1, COS=1, AOS=1, AAOS=1 → 1 |
| (c) | hexahelicene | hexahelicene: POS=0, COS=0, AOS=1, AAOS=0 → 0 |
| (d) | trans-1,2-dimethylcyclohexane (diequatorial chair) | trans-1,2-dimethylcyclohexane (diequatorial chair): POS=0, COS=0, AOS=1, AAOS=0 → 0 |
| (e) | adamantane | adamantane: POS=1, COS=0, AOS=1, AAOS=1 → 0 |
| (f) | cis-1,3-dimethylcyclohexane (diequatorial chair) | cis-1,3-dimethylcyclohexane (diequatorial chair): POS=1, COS=0, AOS=0, AAOS=0 → 0 |
| (g) | cis-1,4-dimethylcyclohexane (chair) | cis-1,4-dimethylcyclohexane (chair): POS=1, COS=0, AOS=0, AAOS=0 → 0 |
| (h) | meso-tartaric acid (anti conformation) | meso-tartaric acid (anti conformation): POS=0, COS=1, AOS=0, AAOS=0 → 0 |
| (i) | gauche-butane | gauche-butane: POS=0, COS=0, AOS=1, AAOS=0 → 0 |
| (j) | anti-1,2-dichloroethane | anti-1,2-dichloroethane: POS=1, COS=1, AOS=1, AAOS=0 → 1 |
| (k) | cyclooctatetraene (tub) | cyclooctatetraene (tub): POS=1, COS=0, AOS=1, AAOS=1 → 0 |

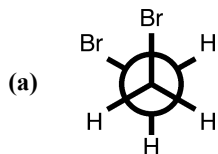
(l) trans-cyclooctene

trans-cyclooctene: POS=0, COS=0, AOS=1, AAOS=0 → 0

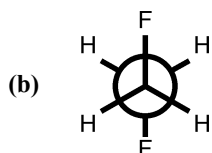
14. Answer: 8

Each Newman projection below shows a staggered conformation of an X-CH₂-CH₂-X molecule (the identical groups X are shown). How many of the conformations are chiral?

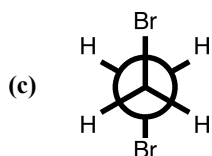
Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.



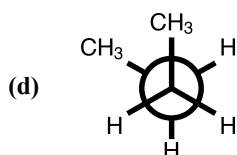
X-CH₂-CH₂-X, g- (C₂: only an axis, chiral conformation) → counts



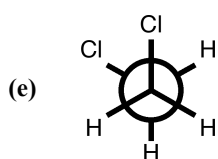
X-CH₂-CH₂-X, anti (C_{2h}: plane, centre, C₂) → does not count



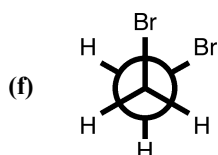
X-CH₂-CH₂-X, anti (C_{2h}: plane, centre, C₂) → does not count



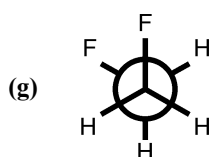
X-CH₂-CH₂-X, g- (C₂: only an axis, chiral conformation) → counts



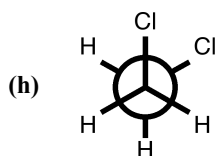
X-CH₂-CH₂-X, g- (C₂: only an axis, chiral conformation) → counts



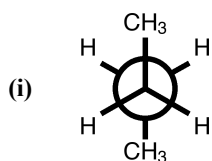
X-CH₂-CH₂-X, g+ (C₂: only an axis, chiral conformation) → counts



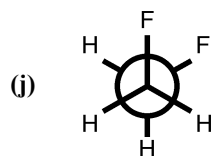
X-CH₂-CH₂-X, g- (C₂: only an axis, chiral conformation) → counts



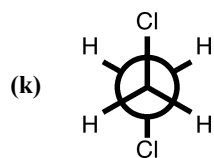
X-CH₂-CH₂-X, g+ (C₂: only an axis, chiral conformation) → counts



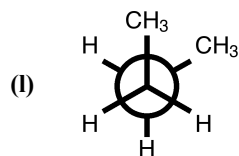
X-CH₂-CH₂-X, anti (C_{2h}: plane, centre, C₂) → does not count



X-CH₂-CH₂-X, g+ (C₂: only an axis, chiral conformation) → counts



X-CH₂-CH₂-X, anti (C_{2h}: plane, centre, C₂) → does not count

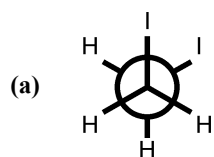


X-CH₂-CH₂-X, g+ (C₂: only an axis, chiral conformation) → counts

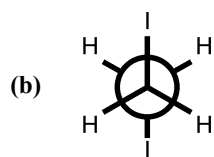
15. Answer: 4

Each Newman projection below shows a staggered conformation of an X-CH₂-CH₂-X molecule. How many of the conformations possess a CENTRE of symmetry?

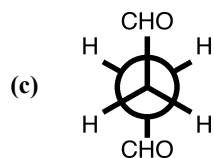
Method: a centre of symmetry (inversion centre $i = S_2$) must send every atom through the centre to an identical atom. In ring compounds this means a group on one face must be matched by the same group on the opposite face of the diametrically opposite carbon, so only rings with an even number of members can have it.



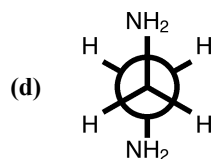
X-CH₂-CH₂-X, g+ (C₂: only an axis, chiral conformation) → does not count



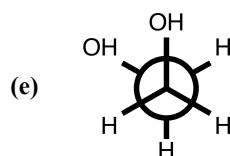
X-CH₂-CH₂-X, anti (C_{2h}: plane, centre, C₂) → counts



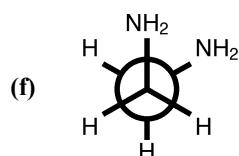
X-CH₂-CH₂-X, anti (C_{2h}: plane, centre, C₂) → counts



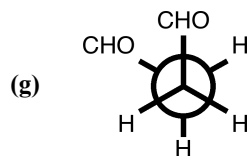
X-CH₂-CH₂-X, anti (C_{2h}: plane, centre, C₂) → counts



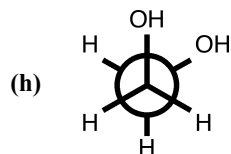
X-CH₂-CH₂-X, g- (C₂: only an axis, chiral conformation) → does not count



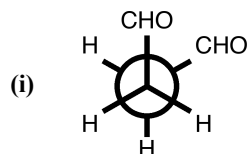
X-CH₂-CH₂-X, g+ (C₂: only an axis, chiral conformation) → does not count



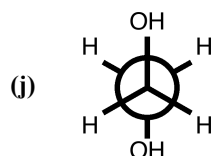
X-CH₂-CH₂-X, g- (C₂: only an axis, chiral conformation) → does not count



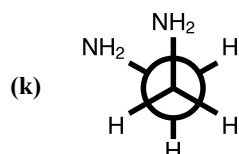
X-CH₂-CH₂-X, g+ (C₂: only an axis, chiral conformation) → does not count



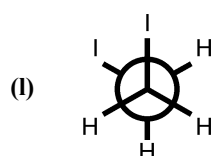
X-CH₂-CH₂-X, g+ (C₂: only an axis, chiral conformation) → does not count



X-CH₂-CH₂-X, anti (C_{2h}: plane, centre, C₂) → counts



X-CH₂-CH₂-X, g- (C₂: only an axis, chiral conformation) → does not count

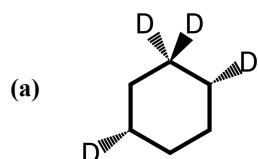


X-CH₂-CH₂-X, g- (C₂: only an axis, chiral conformation) → does not count

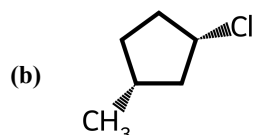
16. Answer: 4

How many of the following have EXACTLY ONE plane of symmetry?

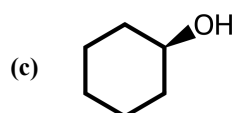
Method: look for a mirror plane that maps the molecule onto itself. In planar-ring drawings the ring plane itself is a plane only if no group is on one face; vertical planes must map every wedge group onto an identical wedge group and every dashed group onto an identical dashed group. Check each structure and count those that have one (or add the planes if the question asks for a total).



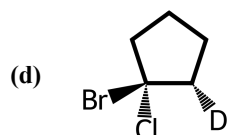
6-membered ring: D on C1 (above the ring), D on C1 (below the ring), D on C3 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



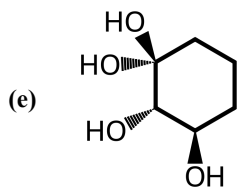
5-membered ring: CH₃ on C3 (below the ring), Cl on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



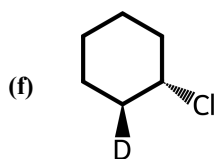
6-membered ring: OH on C6 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → counts



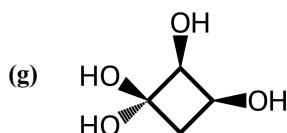
5-membered ring: Br on C3 (above the ring), Cl on C3 (below the ring), D on C4 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



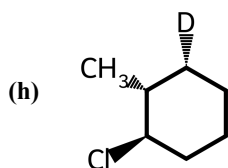
6-membered ring: OH on C2 (above the ring), OH on C2 (below the ring), OH on C3 (below the ring), OH on C4 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



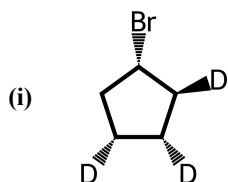
6-membered ring: D on C1 (below the ring), Cl on C2 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



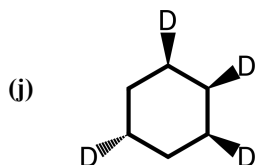
4-membered ring: OH on C1 (above the ring), OH on C2 (above the ring), OH on C2 (below the ring), OH on C4 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



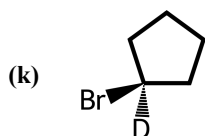
6-membered ring: D on C1 (below the ring), CH₃ on C2 (below the ring), Cl on C3 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



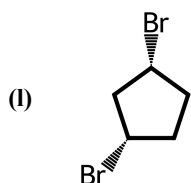
5-membered ring: Br on C1 (below the ring), D on C2 (below the ring), D on C3 (below the ring), D on C4 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



6-membered ring: D on C1 (above the ring), D on C2 (below the ring), D on C3 (below the ring), D on C4 (below the ring), D on C5 (above the ring), D on C6 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → counts



5-membered ring: Br on C1 (below the ring), D on C2 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → counts

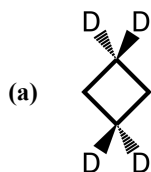


5-membered ring: Br on C1 (below the ring), Br on C3 (below the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → counts

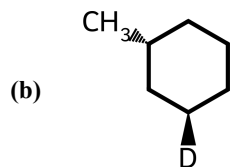
17. Answer: 2

How many of the following have MORE THAN ONE axis of symmetry?

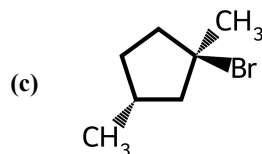
Method: a proper axis C_n rotates the molecule by $360^\circ/n$ into an identical orientation. Check for such rotations (in planar-ring drawings the ring normal and the in-plane axes through opposite atoms or bond mid-points must map every wedge/dash group onto an identical one) and count or add as asked.



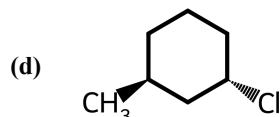
4-membered ring: D on C1 (above the ring), D on C1 (below the ring), D on C3 (above the ring), D on C3 (below the ring). Found: 3 mirror plane(s); an inversion centre (opposite carbons carry the same group on opposite faces); 3 proper axis/axes; no alternate axis; so it is achiral → counts



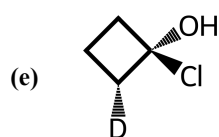
6-membered ring: CH₃ on C2 (below the ring), D on C4 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



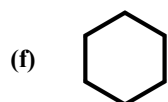
5-membered ring: CH₃ on C3 (below the ring), Br on C5 (above the ring), CH₃ on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



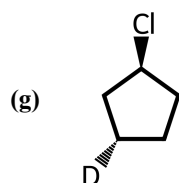
6-membered ring: CH₃ on C3 (above the ring), Cl on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



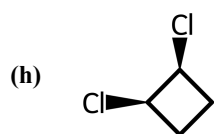
4-membered ring: D on C3 (below the ring), Cl on C4 (above the ring), OH on C4 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



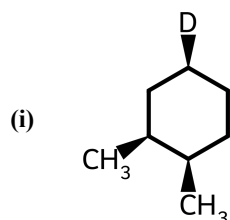
6-membered ring with no substituents. Found: 7 mirror plane(s); an inversion centre (opposite carbons carry the same group on opposite faces); 7 proper axis/axes; an S_n axis (n ≥ 3); so it is achiral → counts



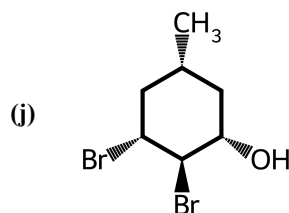
5-membered ring: Cl on C1 (above the ring), D on C3 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



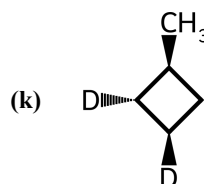
4-membered ring: Cl on C1 (above the ring), Cl on C2 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → does not count



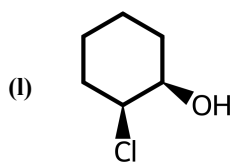
6-membered ring: D on C1 (above the ring), CH₃ on C3 (above the ring), CH₃ on C4 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



6-membered ring: CH₃ on C1 (below the ring), Br on C3 (below the ring), Br on C4 (above the ring), OH on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count



4-membered ring: CH₃ on C1 (above the ring), D on C2 (below the ring), D on C3 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count

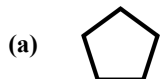


6-membered ring: Cl on C4 (above the ring), OH on C5 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → does not count

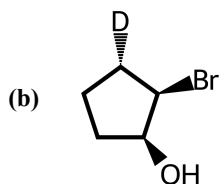
18. Answer: 9

Find the total number of axes of symmetry (every C_n , $n \geq 2$, counted once per axis) present in the compounds (a)–(f) taken together.

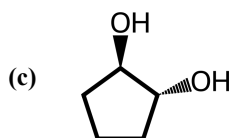
Method: a proper axis C_n rotates the molecule by $360^\circ/n$ into an identical orientation. Check for such rotations (in planar-ring drawings the ring normal and the in-plane axes through opposite atoms or bond mid-points must map every wedge/dash group onto an identical one) and count or add as asked.



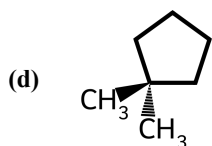
5-membered ring with no substituents. Found: 6 mirror plane(s); no centre of symmetry; 6 proper axis/axes; an S_n axis ($n \geq 3$); so it is achiral → 6



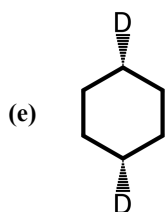
5-membered ring: D on C1 (below the ring), OH on C4 (above the ring), Br on C5 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → 0



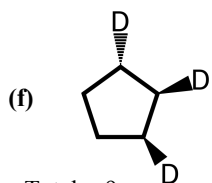
5-membered ring: OH on C1 (above the ring), OH on C5 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → 1



5-membered ring: CH_3 on C3 (above the ring), CH_3 on C3 (below the ring). Found: 2 mirror plane(s); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is achiral → 1



6-membered ring: D on C1 (below the ring), D on C4 (below the ring). Found: 2 mirror plane(s); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is achiral → 1



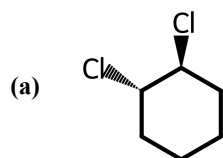
5-membered ring: D on C1 (below the ring), D on C4 (above the ring), D on C5 (above the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; no proper axis; no alternate axis; so it is chiral → 0

Total = 9

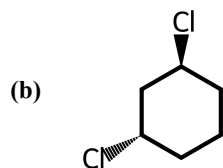
19. Answer: 10

Mark POS (plane), COS (centre), AOS (axis, C_n with $n \geq 2$) and AAOS (alternate axis, S_n with $n \geq 3$) for each of the following compounds. Find the total number of marks.

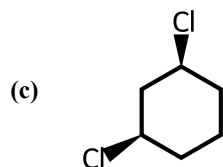
Method: list the symmetry elements of every structure. A plane of symmetry (POS) divides the molecule into two mirror-image halves; a centre of symmetry (COS) maps every atom to an identical atom through the centre; an axis of symmetry (AOS) is a proper rotation axis C_n ($n \geq 2$) after which the molecule looks unchanged; an alternate axis (AAOS) is a rotation–reflection axis S_n ($n \geq 3$). Rings are taken as planar with wedge groups above and dashed groups below the ring plane; for each structure decide which of the four elements exist, then add up the marks.



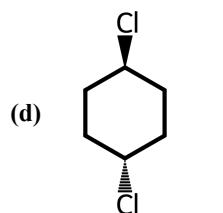
trans-1,2-dichlorocyclohexane. 6-membered ring: Cl on C1 (above the ring), Cl on C2 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → 1



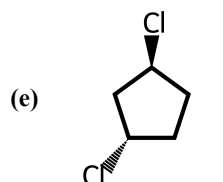
trans-1,3-dichlorocyclohexane. 6-membered ring: Cl on C1 (above the ring), Cl on C3 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → 1



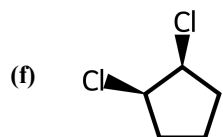
cis-1,3-dichlorocyclohexane. 6-membered ring: Cl on C1 (above the ring), Cl on C3 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → 1



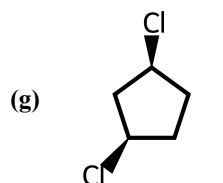
trans-1,4-dichlorocyclohexane. 6-membered ring: Cl on C1 (above the ring), Cl on C4 (below the ring). Found: 1 mirror plane(s); an inversion centre (opposite carbons carry the same group on opposite faces); 1 proper axis/axes; no alternate axis; so it is achiral → 3



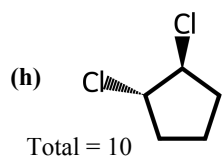
trans-1,3-dichlorocyclopentane. 5-membered ring: Cl on C1 (above the ring), Cl on C3 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → 1



cis-1,2-dichlorocyclopentane. 5-membered ring: Cl on C1 (above the ring), Cl on C2 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → 1



cis-1,3-dichlorocyclopentane. 5-membered ring: Cl on C1 (above the ring), Cl on C3 (above the ring). Found: 1 mirror plane(s); no centre of symmetry; no proper axis; no alternate axis; so it is achiral → 1



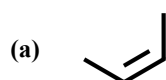
trans-1,2-dichlorocyclopentane. 5-membered ring: Cl on C1 (above the ring), Cl on C2 (below the ring). Found: no mirror plane (no reflection maps every group onto an identical group on the same face); no centre of symmetry; 1 proper axis/axes; no alternate axis; so it is chiral → 1

Total = 10

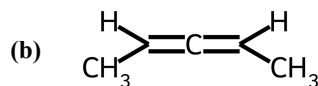
20. Answer: 13

Mark POS (plane), COS (centre), AOS (axis, C_n with $n \geq 2$) and AAOS (alternate axis, S_n with $n \geq 3$) for each of the following alkenes and cumulenes (planar cis/trans isomers for an odd number of cumulated double bonds, axial chirality for an even number). Find the total number of marks.

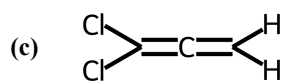
Method: list the symmetry elements of every structure. A plane of symmetry (POS) divides the molecule into two mirror-image halves; a centre of symmetry (COS) maps every atom to an identical atom through the centre; an axis of symmetry (AOS) is a proper rotation axis C_n ($n \geq 2$) after which the molecule looks unchanged; an alternate axis (AAOS) is a rotation-reflection axis S_n ($n \geq 3$). Rings are taken as planar with wedge groups above and dashed groups below the ring plane; for each structure decide which of the four elements exist, then add up the marks.



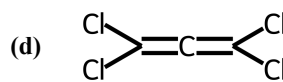
(Z)-but-2-ene: POS=1, COS=0, AOS=1, AAOS=0



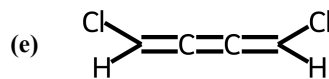
penta-2,3-diene: POS=0, COS=0, AOS=1, AAOS=0



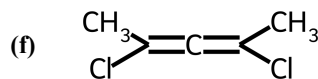
1,1-dichloroallene: POS=1, COS=0, AOS=1, AAOS=0



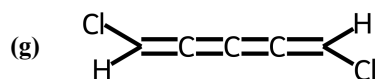
1,1,3,3-tetrachloroallene: POS=1, COS=0, AOS=1, AAOS=1



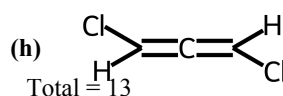
cis-1,4-dichlorobuta-1,2,3-triene: POS=1, COS=0, AOS=1, AAOS=0



1,3-dimethyl-1,3-dichloroallene: POS=0, COS=0, AOS=1, AAOS=0



1,5-dichloropenta-1,2,3,4-tetraene: POS=0, COS=0, AOS=1, AAOS=0



1,3-dichloroallene: POS=0, COS=0, AOS=1, AAOS=0

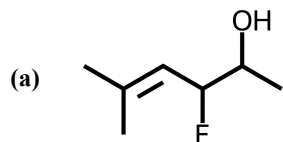
Total = 13

Part B — Identification of stereo-features (count the compounds)

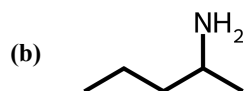
21. Answer: 9

How many of the following compounds can show optical isomerism?

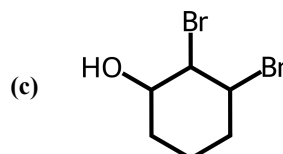
Method: a compound can show optical isomerism if at least one stereoisomer of the skeleton drawn is chiral. Mark every stereogenic centre (an sp^3 atom with four different groups) and any chiral axis, write the possible stereoisomers and test them for a plane, centre or alternate axis of symmetry. The structures are drawn flat, so all possible stereoisomers must be considered.



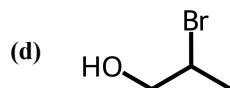
has at least one optically active stereoisomer → counts



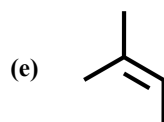
has at least one optically active stereoisomer → counts



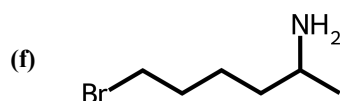
has at least one optically active stereoisomer → counts



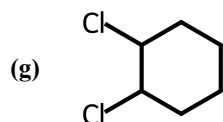
2-bromopropan-1-ol: has at least one optically active stereoisomer → counts



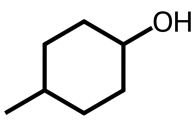
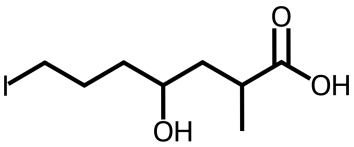
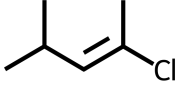
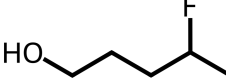
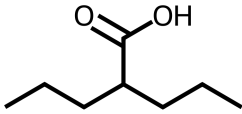
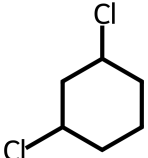
no chiral stereoisomer exists → does not count



has at least one optically active stereoisomer → counts



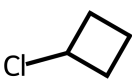
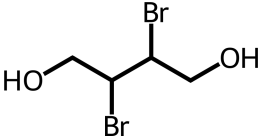
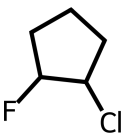
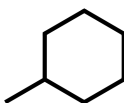
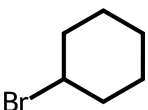
1,2-dichlorocyclohexane: has at least one optically active stereoisomer → counts

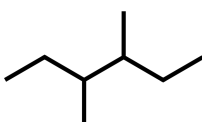
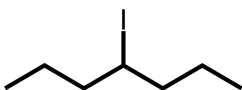
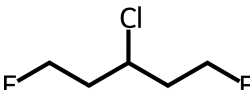
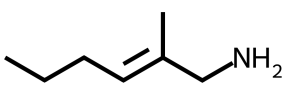
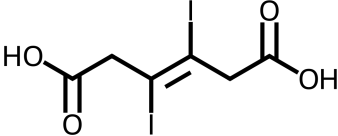
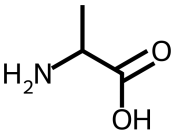
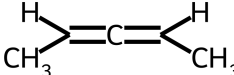
- (h)  4-methylcyclohexanol: no chiral stereoisomer exists → does not count
- (i)  no chiral stereoisomer exists → does not count
- (j)  has at least one optically active stereoisomer → counts
- (k)  no chiral stereoisomer exists → does not count
- (l)  has at least one optically active stereoisomer → counts
- (m)  no chiral stereoisomer exists → does not count
- (n)  1,3-dichlorocyclohexane: has at least one optically active stereoisomer → counts

22. Answer: 9

How many of the following compounds can show stereoisomerism (geometrical and/or optical)?

Method: a compound shows stereoisomerism if it has a stereogenic centre, a double bond (or ring) with two different groups on each carbon, or a chiral axis/plane. Check each structure for any of these.

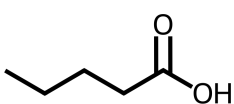
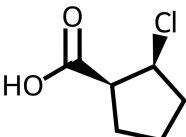
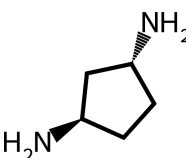
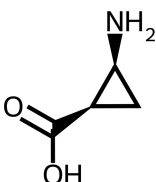
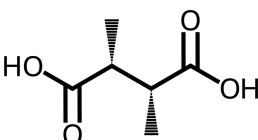
- (a)  1 stereoisomer(s) possible → does not count
- (b)  2,3-dibromobutane-1,4-diol: 3 stereoisomer(s) possible → counts
- (c)  4 stereoisomer(s) possible → counts
- (d)  trans-cyclooctene: 2 stereoisomer(s) possible → counts
- (e)  methylcyclohexane: 1 stereoisomer(s) possible → does not count
- (f)  1 stereoisomer(s) possible → does not count

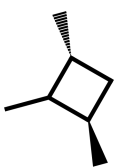
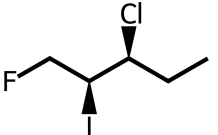
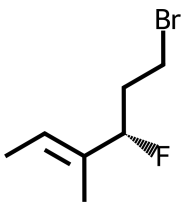
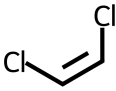
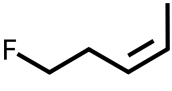
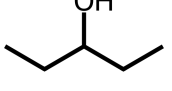
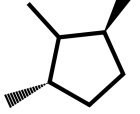
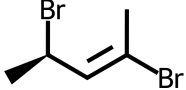
- (g)  3 stereoisomer(s) possible → counts
- (h)  1 stereoisomer(s) possible → does not count
- (i)  1 stereoisomer(s) possible → does not count
- (j)  2 stereoisomer(s) possible → counts
- (k)  but-2-ene: 2 stereoisomer(s) possible → counts
- (l)  2 stereoisomer(s) possible → counts
- (m)  alanine: 2 stereoisomer(s) possible → counts
- (n)  penta-2,3-diene: 2 stereoisomer(s) possible → counts

23. Answer: 9

How many of the following molecules, as drawn, are optically active (chiral and a single pure stereoisomer)?

Method: here the exact stereoisomer drawn is considered. It is optically active when it has no plane, centre or alternate axis of symmetry; meso forms and compounds without stereocentres are inactive.

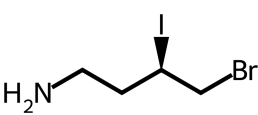
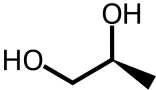
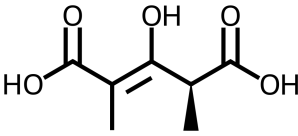
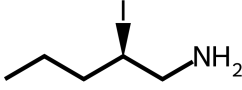
- (a)  achiral → does not count
- (b)  chiral (no plane/centre of symmetry) → counts
- (c)  chiral (no plane/centre of symmetry) → counts
- (d)  chiral (no plane/centre of symmetry) → counts
- (e)  chiral (no plane/centre of symmetry) → counts

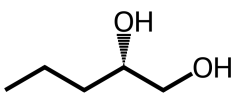
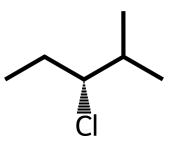
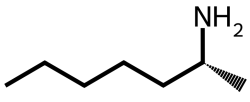
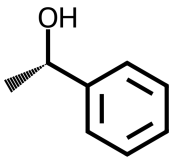
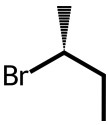
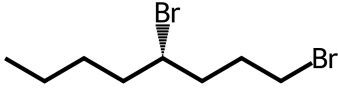
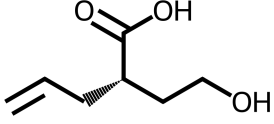
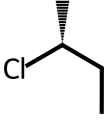
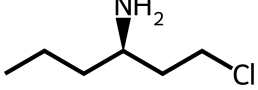
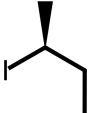
- (f)  chiral (no plane/centre of symmetry) → counts
- (g)  achiral → does not count
- (h)  chiral (no plane/centre of symmetry) → counts
- (i)  chiral (no plane/centre of symmetry) → counts
- (j)  1,2-dichloroethene: achiral → does not count
- (k)  achiral → does not count
- (l)  3-pentanol: achiral → does not count
- (m)  chiral (no plane/centre of symmetry) → counts
- (n)  chiral (no plane/centre of symmetry) → counts

24. Answer: 7

How many of the following compounds have 'R' configuration at the stereocentre (Cahn–Ingold–Prelog)?

Method (CIP rules): rank the four groups by atomic number of the atom directly attached; on a tie go outwards to the next atoms (duplicate atoms for double bonds). View with the lowest-priority group pointing away; 1→2→3 clockwise is R, anticlockwise is S. For a wedge/dash drawing swap parity if the lowest priority group points towards you.

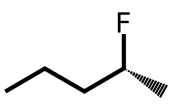
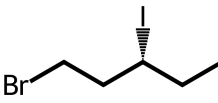
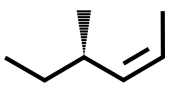
- (a)  R
- (b)  propane-1,2-diol: S
- (c)  S
- (d)  R

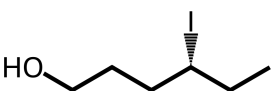
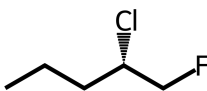
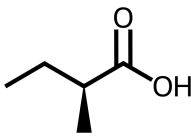
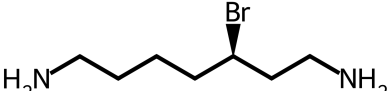
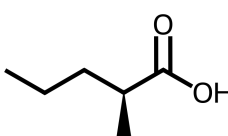
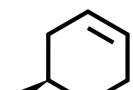
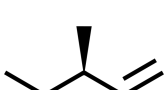
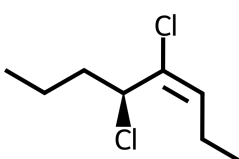
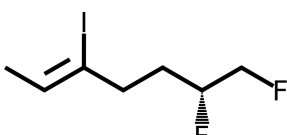
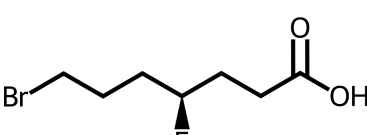
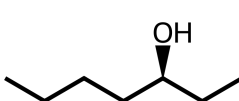
- (e)  S
- (f)  3-chloro-2-methylpentane: R
- (g)  R
- (h)  1-phenylethanol: S
- (i)  2-bromobutane: R
- (j)  S
- (k)  S
- (l)  2-chlorobutane: R
- (m)  R
- (n)  2-iodobutane: S

25. Answer: 6

How many of the following compounds have 'S' configuration at the stereocentre (Cahn–Ingold–Prelog)?

Method (CIP rules): rank the four groups by atomic number of the atom directly attached; on a tie go outwards to the next atoms (duplicate atoms for double bonds). View with the lowest-priority group pointing away; 1→2→3 clockwise is R, anticlockwise is S. For a wedge/dash drawing swap parity if the lowest priority group points towards you.

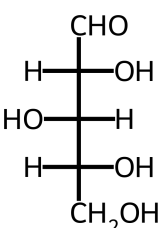
- (a)  R
- (b)  R
- (c)  4-methylhex-2-ene: S

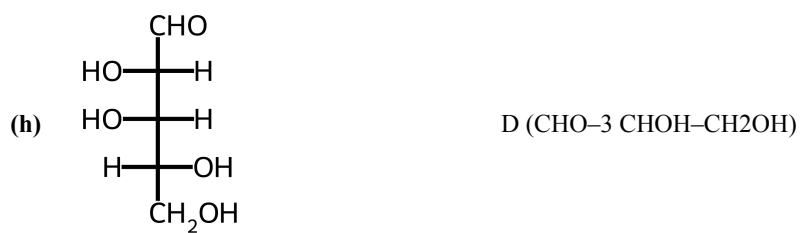
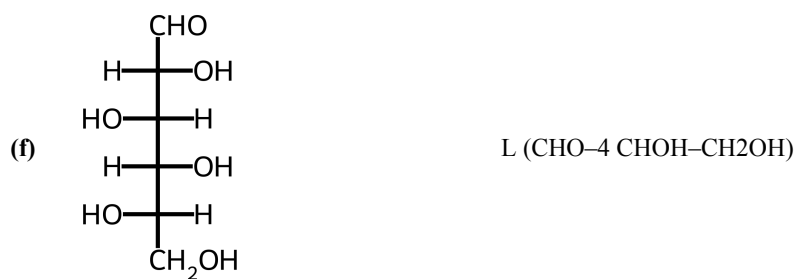
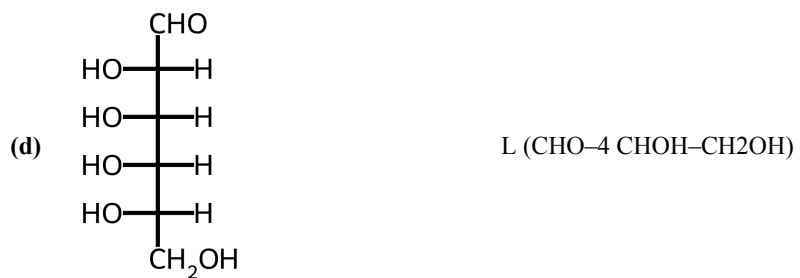
- (d)  R
- (e)  S
- (f)  S
- (g)  R
- (h)  S
- (i)  cyclohexene-4-methyl: R
- (j)  3-methylpent-1-ene: R
- (k)  S
- (l)  R
- (m)  R
- (n)  S

26. Answer: 7

How many of the following Fischer projections represent D-configuration (reference: the stereocentre farthest from the top (most oxidised) carbon; -OH/-NH₂ on the right side)?

Method: in a Fischer projection with the most oxidised carbon (CHO or COOH) at the top, look at the stereocentre farthest from the top. -OH (or -NH₂) on the right means D, on the left means L. Newman, sawhorse and wedge drawings must first be converted to a Fischer projection.

- (a)  D (CHO-3 CHOH-CH₂OH)



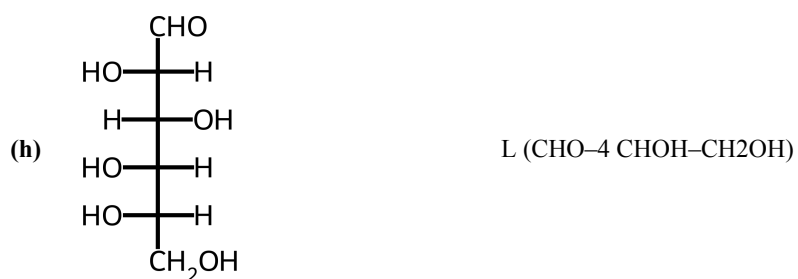
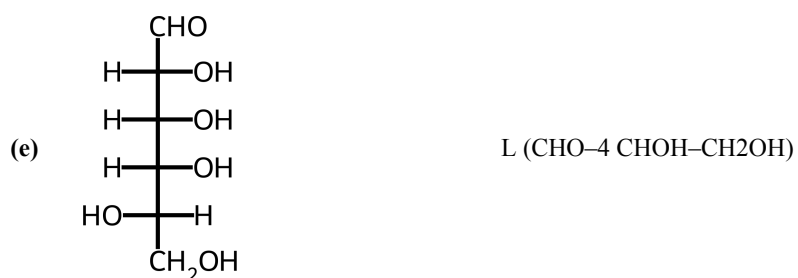


27. Answer: 7

How many of the following Fischer projections represent L-configuration (reference: the stereocentre farthest from the top (most oxidised) carbon; -OH/-NH₂ on the left side)?

Method: in a Fischer projection with the most oxidised carbon (CHO or COOH) at the top, look at the stereocentre farthest from the top. -OH (or -NH₂) on the right means D, on the left means L. Newman, sawhorse and wedge drawings must first be converted to a Fischer projection.



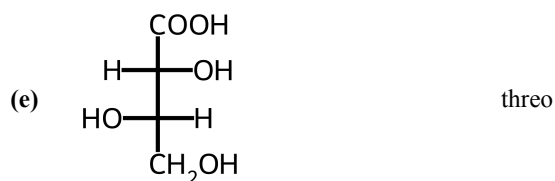
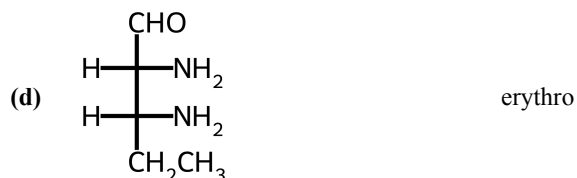
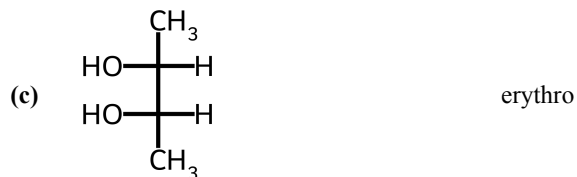
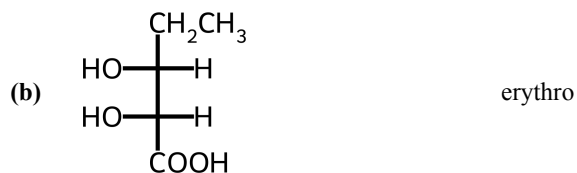
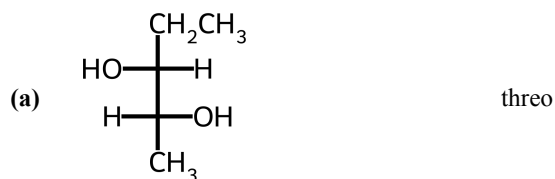




28. Answer: 6

How many of the following Fischer projections are ERYTHRO isomers? (Erythro: the two similar (non-hydrogen) substituents on the two adjacent stereocentres lie on the SAME side of the vertical chain.)

Method: convert each structure to a Fischer projection. If the two similar (non-hydrogen) substituents of the adjacent stereocentres are on the same side the compound is erythro; on opposite sides it is threo.



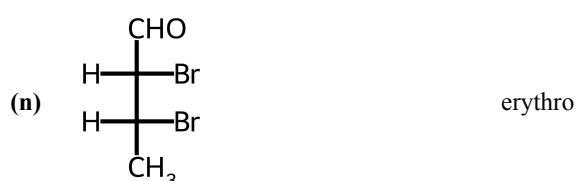
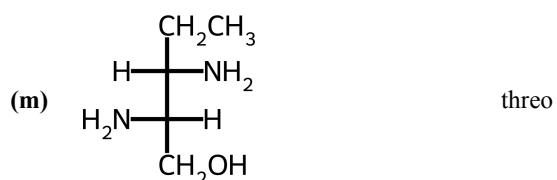
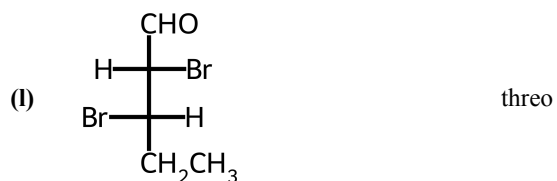
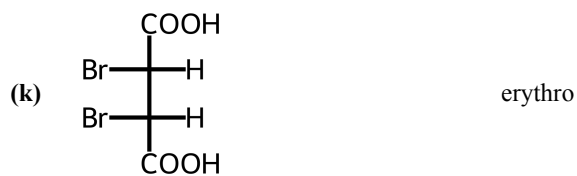
(f)	$ \begin{array}{c} \text{CH}_2\text{CH}_3 \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{CH}_2\text{OH} \end{array} $	threo
(g)	$ \begin{array}{c} \text{CH}_2\text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{Cl} - \text{C} - \text{H} \\ \\ \text{COOH} \end{array} $	threo
(h)	$ \begin{array}{c} \text{CHO} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{CH}_2\text{CH}_3 \end{array} $	erythro
(i)	$ \begin{array}{c} \text{CHO} \\ \\ \text{Cl} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{CH}_2\text{CH}_3 \end{array} $	threo
(j)	$ \begin{array}{c} \text{COOH} \\ \\ \text{Cl} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{COOH} \end{array} $	threo
(k)	$ \begin{array}{c} \text{COOH} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{CH}_2\text{OH} \end{array} $	erythro
(l)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{CH}_2\text{CH}_3 \end{array} $	erythro
(m)	$ \begin{array}{c} \text{CHO} \\ \\ \text{Cl} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{CH}_2\text{OH} \end{array} $	threo
(n)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{Br} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{CH}_3 \end{array} $	threo

29. Answer: 8

How many of the following Fischer projections are THREO isomers? (Threo: the two similar (non-hydrogen) substituents on the two adjacent stereocentres lie on OPPOSITE sides of the vertical chain.)

Method: convert each structure to a Fischer projection. If the two similar (non-hydrogen) substituents of the adjacent stereocentres are on the same side the compound is erythro; on opposite sides it is threo.

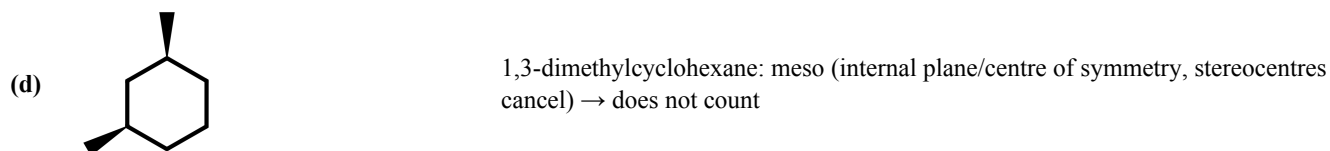
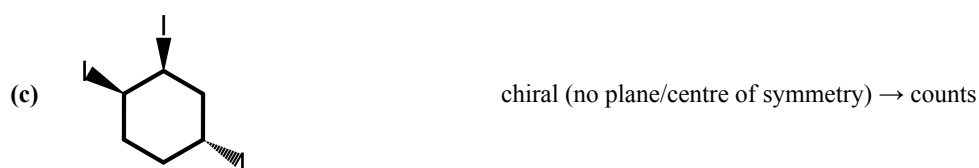
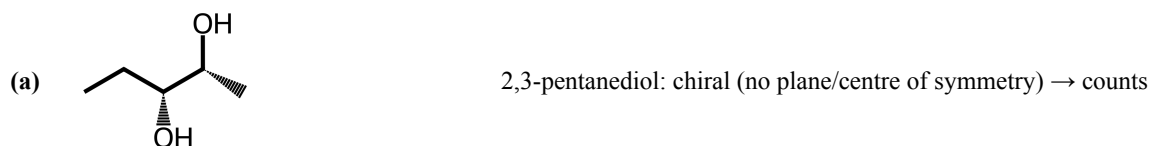
(a)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{Cl} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{COOH} \end{array} $	threo
(b)	$ \begin{array}{c} \text{COOH} \\ \\ \text{H}_2\text{N} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{NH}_2 \\ \\ \text{CH}_2\text{CH}_3 \end{array} $	threo
(c)	$ \begin{array}{c} \text{CH}_2\text{CH}_3 \\ \\ \text{H}_2\text{N} - \text{C} - \text{H} \\ \\ \text{H}_2\text{N} - \text{C} - \text{H} \\ \\ \text{COOH} \end{array} $	erythro
(d)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{CH}_2\text{CH}_3 \end{array} $	erythro
(e)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{Br} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{CH}_2\text{OH} \end{array} $	threo
(f)	$ \begin{array}{c} \text{COOH} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{COOH} \end{array} $	erythro
(g)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{Cl} - \text{C} - \text{H} \\ \\ \text{Cl} - \text{C} - \text{H} \\ \\ \text{CH}_3 \end{array} $	erythro
(h)	$ \begin{array}{c} \text{CH}_2\text{CH}_3 \\ \\ \text{H} - \text{C} - \text{NH}_2 \\ \\ \text{H}_2\text{N} - \text{C} - \text{H} \\ \\ \text{CH}_3 \end{array} $	threo
(i)	$ \begin{array}{c} \text{CH}_2\text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{Br} - \text{C} - \text{H} \\ \\ \text{COOH} \end{array} $	threo
(j)	$ \begin{array}{c} \text{CH}_2\text{CH}_3 \\ \\ \text{H}_2\text{N} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{NH}_2 \\ \\ \text{CH}_2\text{CH}_3 \end{array} $	threo

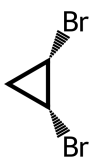
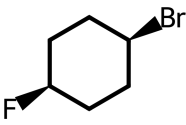
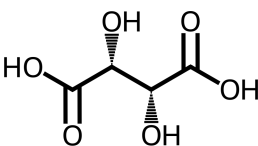
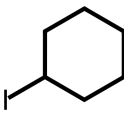
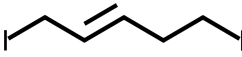
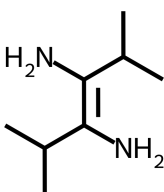
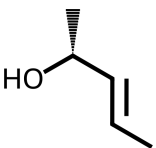
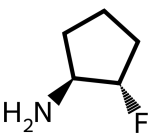
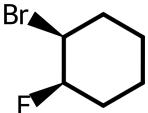


30. Answer: 6

How many of the following compounds can be resolved into enantiomers at room temperature (stable, separable enantiomers exist for the drawn species)?

Method: a compound can be resolved if it is chiral and its enantiomers are configurationally stable at room temperature (no fast inversion/rotation). Species with a pyramidal N of three different groups invert rapidly and cannot be resolved.

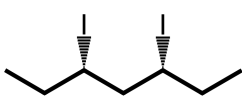
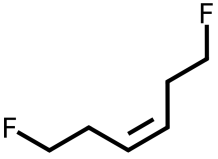
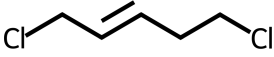


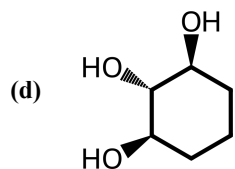
- (f)  meso (internal plane/centre of symmetry, stereocentres cancel) → does not count
- (g)  achiral: cis/trans-type isomer with plane of symmetry, no chiral centre → does not count
- (h)  tartaric acid: chiral (no plane/centre of symmetry) → counts
- (i)  achiral → does not count
- (j)  achiral → does not count
- (k)  achiral → does not count
- (l)  chiral (no plane/centre of symmetry) → counts
- (m)  chiral (no plane/centre of symmetry) → counts
- (n)  chiral (no plane/centre of symmetry) → counts

31. Answer: 5

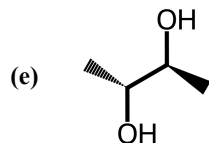
How many of the following molecules are meso compounds?

Method: a meso compound has two or more stereocentres but is achiral because of an internal plane or centre of symmetry. Look for stereocentres arranged symmetrically with opposite configurations.

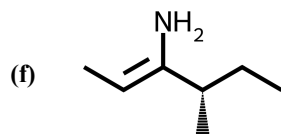
- (a)  meso (internal plane/centre of symmetry, stereocentres cancel) → counts
- (b)  achiral → does not count
- (c)  achiral → does not count



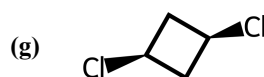
cyclohexane-1,2,3-triol: meso (internal plane/centre of symmetry, stereocentres cancel) → counts



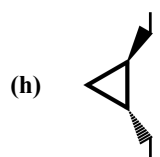
2,3-butanediol: meso (internal plane/centre of symmetry, stereocentres cancel) → counts



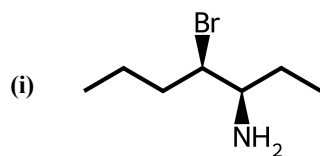
chiral (no plane/centre of symmetry) → does not count



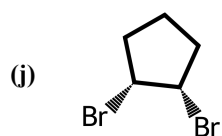
1,3-dichlorocyclobutane: achiral: cis/trans-type isomer with plane of symmetry, no chiral centre → does not count



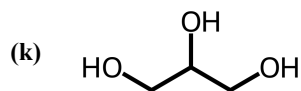
chiral (no plane/centre of symmetry) → does not count



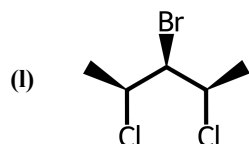
chiral (no plane/centre of symmetry) → does not count



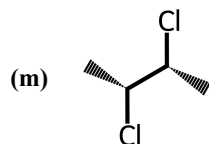
meso (internal plane/centre of symmetry, stereocentres cancel) → counts



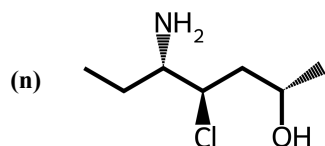
glycerol: achiral → does not count



meso (internal plane/centre of symmetry, stereocentres cancel) → counts



2,3-dichlorobutane: chiral (no plane/centre of symmetry) → does not count



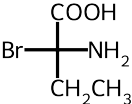
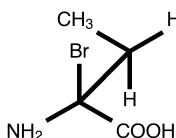
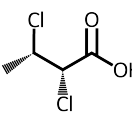
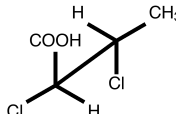
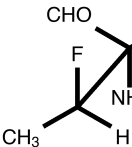
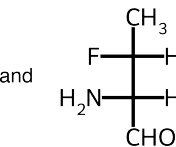
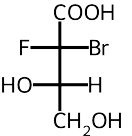
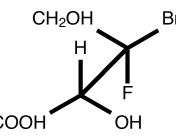
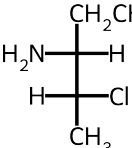
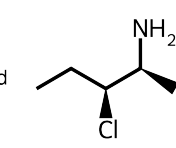
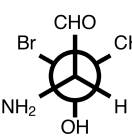
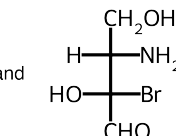
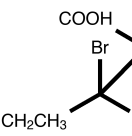
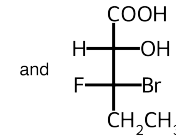
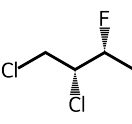
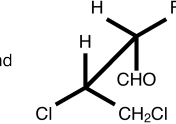
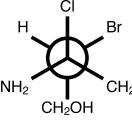
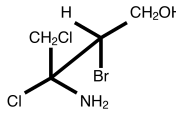
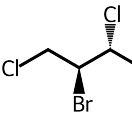
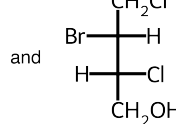
chiral (no plane/centre of symmetry) → does not count

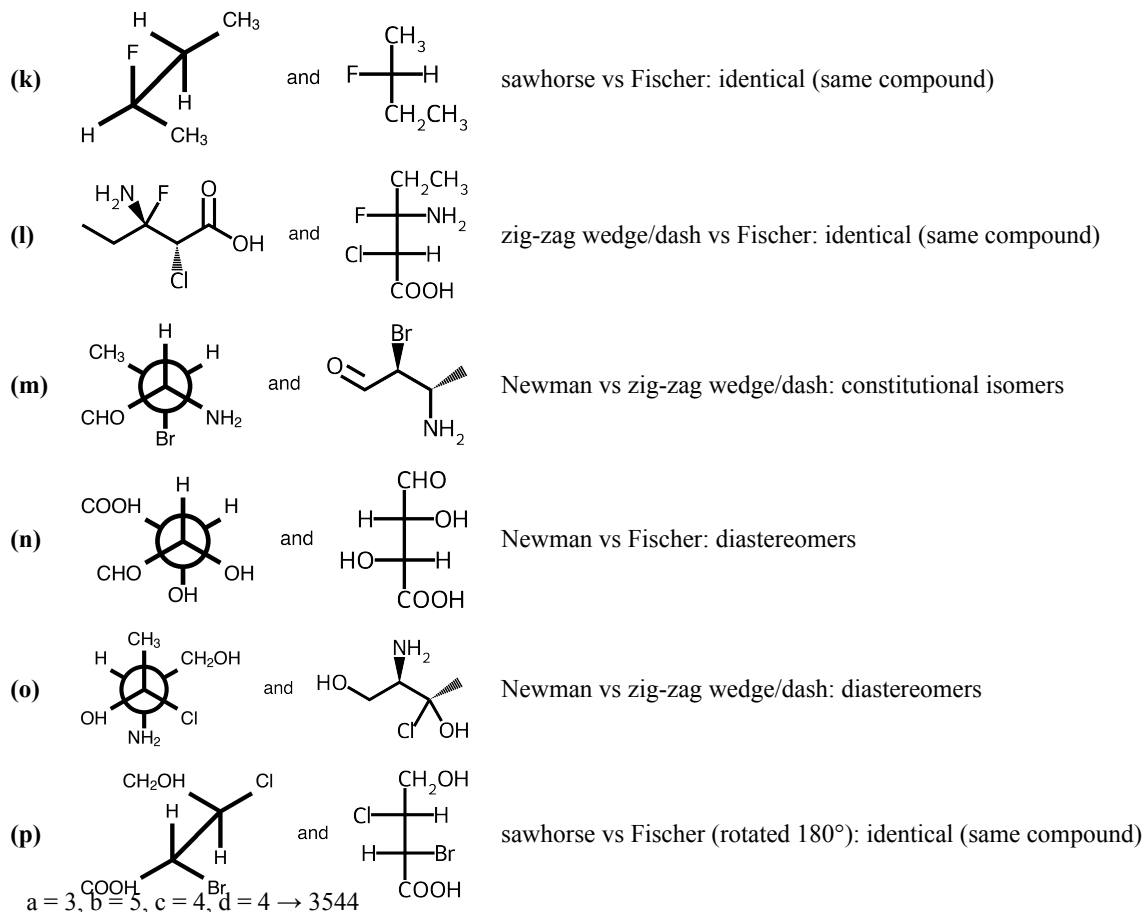
Part C — Relationship between pairs (all projection formulas)

32. Answer: 3544

Each of the following pairs is drawn in two different projection formulas (Fischer, zig-zag wedge/dash, Newman, sawhorse). A Fischer projection may be turned only through 180° in the plane; Newman and sawhorse conformations may differ. Let a, b, c and d be the numbers of pairs that are ENANTIOMERS, DIASTEREOMERS, IDENTICAL molecules (homomers) and CONSTITUTIONAL isomers respectively. Find the four-digit number abcd.

Method: bring both members of each pair to one common representation (e.g. a Fischer projection; rotate Fischer projections only by 180° in the plane; Newman and sawhorse conformations are interconverted by rotation about the C–C bond). Then compare: different connectivity $\rightarrow$ constitutional isomers; same connectivity and same configuration at every centre $\rightarrow$ identical; mirror-image configuration at every centre $\rightarrow$ enantiomers; anything else $\rightarrow$ diastereomers. Count a, b, c, d.

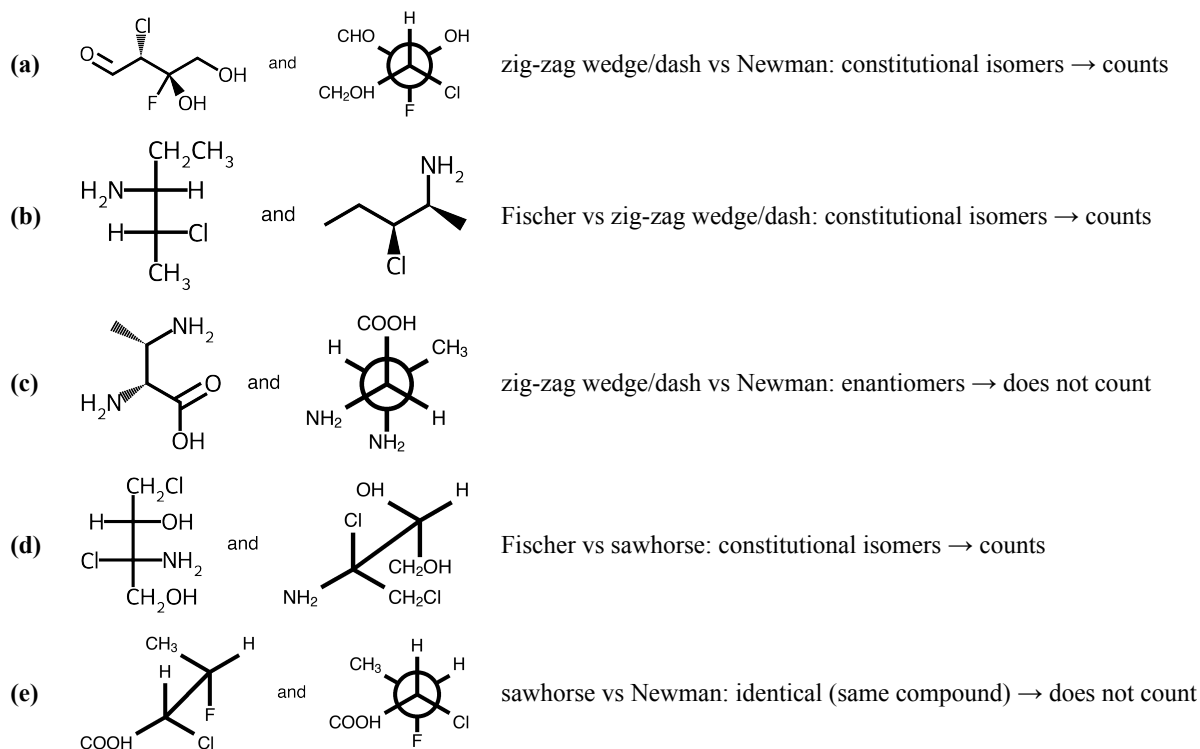
- (a)  and  Fischer vs sawhorse: identical (same compound)
- (b)  and  zig-zag wedge/dash vs sawhorse: enantiomers
- (c)  and  sawhorse vs Fischer: diastereomers
- (d)  and  Fischer vs sawhorse: constitutional isomers
- (e)  and  Fischer vs zig-zag wedge/dash: constitutional isomers
- (f)  and  Newman vs Fischer (rotated 180°): constitutional isomers
- (g)  and  sawhorse vs Fischer (rotated 180°): enantiomers
- (h)  and  zig-zag wedge/dash vs sawhorse: enantiomers
- (i)  and  Newman vs sawhorse: diastereomers
- (j)  and  zig-zag wedge/dash vs Fischer (rotated 180°): diastereomers

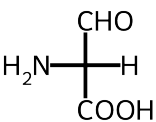
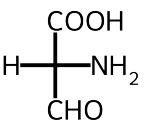
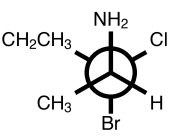
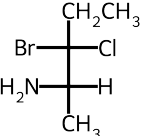
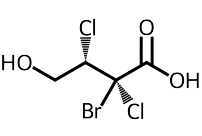
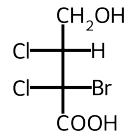
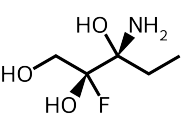
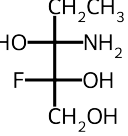
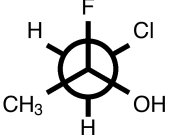
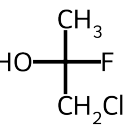
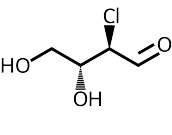
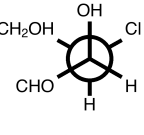
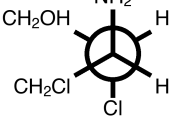
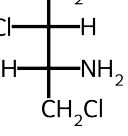


33. Answer: 6

How many of the following pairs can, in principle, be separated by simple fractional distillation? (Assume that compounds with different physical constants in an achiral medium can be separated.)

Method: enantiomers have the same boiling point and are not separated by distillation, whereas diastereomers and constitutional isomers are. So each racemic pair and each achiral stereoisomer forms one fraction. Chiral stereoisomers are optically active and obtained by a chiral column from the racemic fractions.

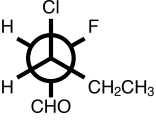
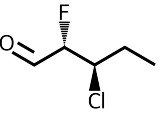
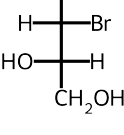
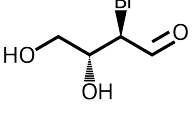
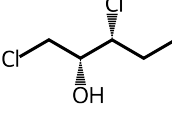
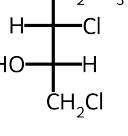
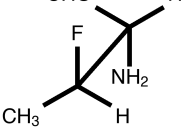
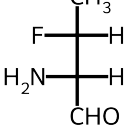


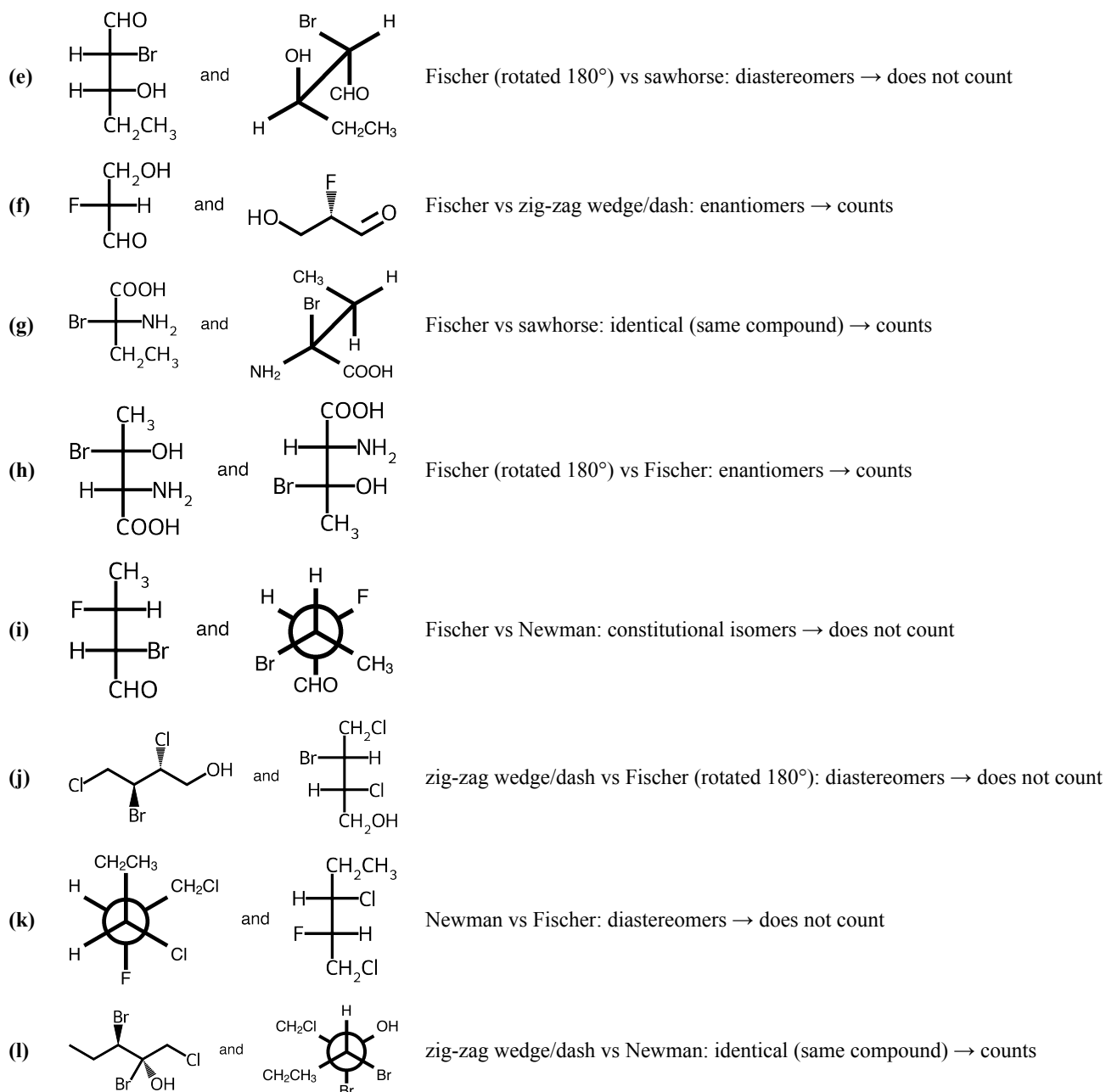
- (f)  and  Fischer (rotated 180°) vs Fischer: identical (same compound) → does not count
- (g)  and  Newman vs Fischer (rotated 180°): diastereomers → counts
- (h)  and  zig-zag wedge/dash vs Fischer (rotated 180°): enantiomers → does not count
- (i)  and  zig-zag wedge/dash vs Fischer (rotated 180°): diastereomers → counts
- (j)  and  Newman vs Fischer (rotated 180°): identical (same compound) → does not count
- (k)  and  zig-zag wedge/dash vs Newman: constitutional isomers → counts
- (l)  and  Newman vs Fischer (rotated 180°): enantiomers → does not count

34. Answer: 6

In how many of the following pairs do both members necessarily have the same boiling point, melting point, vapour pressure, density and solubility in an achiral solvent?

Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.

- (a)  and  Newman vs zig-zag wedge/dash: identical (same compound) → counts
- (b)  and  Fischer (rotated 180°) vs zig-zag wedge/dash: diastereomers → does not count
- (c)  and  zig-zag wedge/dash vs Fischer (rotated 180°): enantiomers → counts
- (d)  and  sawhorse vs Fischer: diastereomers → does not count



35. Answer: 4

The four stereoisomers I–IV of the same compound are drawn below. Among the six possible pairs (each unordered pair counted once), how many are pairs of **DIASTEREOMERS**?

Method: convert both members to the same representation and compare configuration at every stereocentre: all centres identical → same compound (homomers); all centres opposite → enantiomers; some same and some opposite → diastereomers; different connectivity → constitutional isomers. Then apply the property asked (enantiomers share all physical properties except the sign of rotation; diastereomers and constitutional isomers differ).

- I–II: diastereomers
- I–III: diastereomers
- I–IV: enantiomers
- II–III: enantiomers
- II–IV: diastereomers
- III–IV: diastereomers

→ 4

36. Answer: 1

The four stereoisomers I–IV of the same compound are drawn below. Among the six possible pairs (each unordered pair counted once), how many are pairs of ENANTIOMERS?

Method: convert both members to the same representation and compare configuration at every stereocentre: all centres identical → same compound (homomers); all centres opposite → enantiomers; some same and some opposite → diastereomers; different connectivity → constitutional isomers. Then apply the property asked (enantiomers share all physical properties except the sign of rotation; diastereomers and constitutional isomers differ).

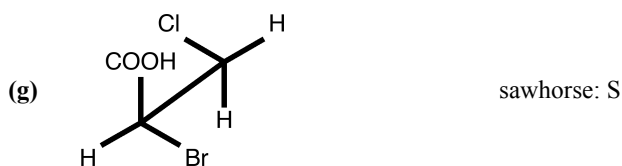
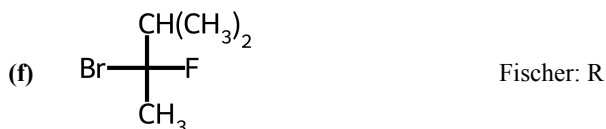
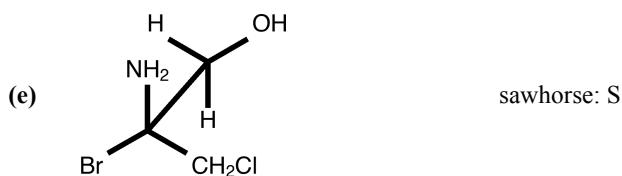
I–II: diastereomers
 I–III: diastereomers
 I–IV: enantiomers
 II–III: diastereomers
 II–IV: diastereomers
 III–IV: diastereomers
 → 1

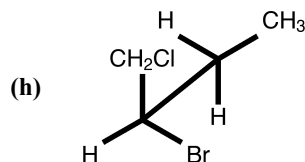
Part D — R/S, D/L and erythro/threo in different projections

37. Answer: 7

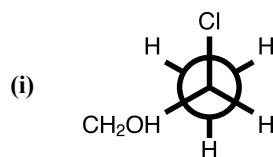
How many of the following compounds, drawn in different projection formulas, have 'R' configuration at the stereogenic carbon? (Newman and sawhorse show the stereogenic carbon in front.)

Method (CIP rules): rank the four groups by atomic number of the atom directly attached; on a tie go outwards to the next atoms (duplicate atoms for double bonds). View with the lowest-priority group pointing away; 1→2→3 clockwise is R, anticlockwise is S. For a wedge/dash drawing swap parity if the lowest priority group points towards you.

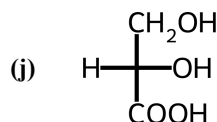




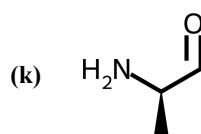
sawhorse: R



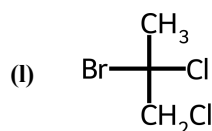
Newman: R



Fischer: S



zig-zag wedge/dash: R

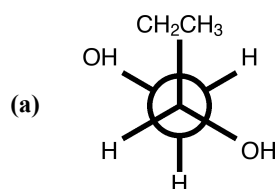


Fischer (rotated 180°): S

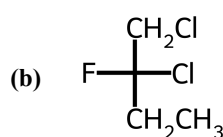
38. Answer: 8

How many of the following compounds, drawn in different projection formulas, have 'S' configuration at the stereogenic carbon? (Newman and sawhorse show the stereogenic carbon in front.)

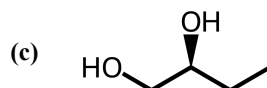
Method (CIP rules): rank the four groups by atomic number of the atom directly attached; on a tie go outwards to the next atoms (duplicate atoms for double bonds). View with the lowest-priority group pointing away; 1→2→3 clockwise is R, anticlockwise is S. For a wedge/dash drawing swap parity if the lowest priority group points towards you.



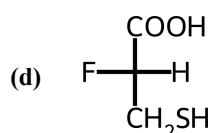
Newman: S



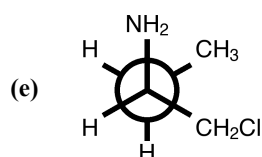
Fischer: S



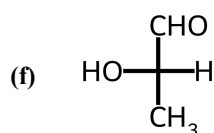
zig-zag wedge/dash: S



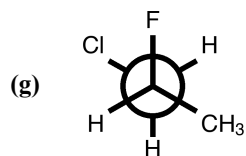
Fischer (rotated 180°): R



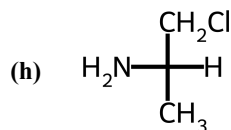
Newman: S



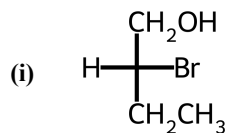
Fischer (rotated 180°): S



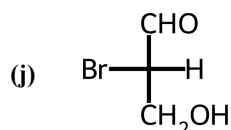
Newman: R



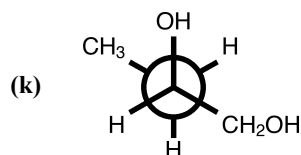
Fischer: S



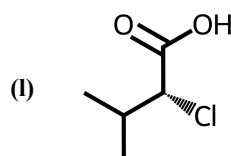
Fischer (rotated 180°): R



Fischer (rotated 180°): S



Newman: S

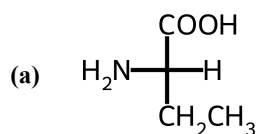


zig-zag wedge/dash: R

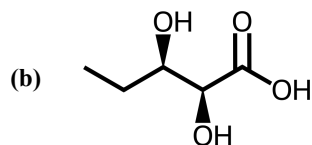
39. Answer: 7

How many of the following compounds, drawn in different projection formulas, have D-configuration? (Convert mentally to a Fischer projection with the most oxidised carbon, $-\text{CHO}$ or $-\text{COOH}$, at the top; the D/L label is decided by the stereocentre farthest from it.)

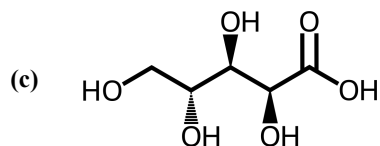
Method: in a Fischer projection with the most oxidised carbon (CHO or COOH) at the top, look at the stereocentre farthest from the top. $-\text{OH}$ (or $-\text{NH}_2$) on the right means D, on the left means L. Newman, sawhorse and wedge drawings must first be converted to a Fischer projection.



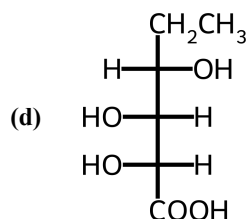
Fischer: L



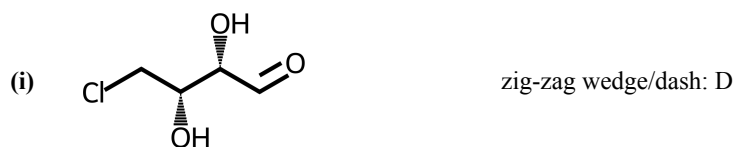
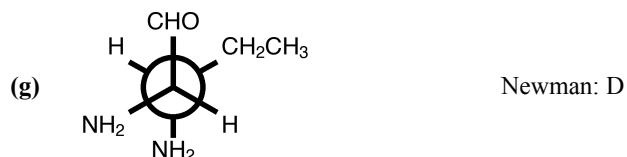
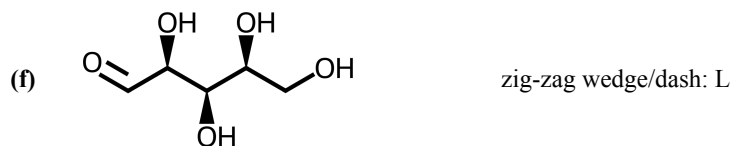
zig-zag wedge/dash: D



zig-zag wedge/dash: D



Fischer (rotated 180°): L

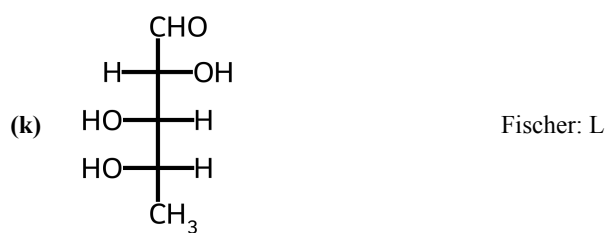
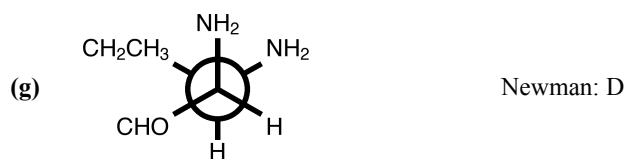
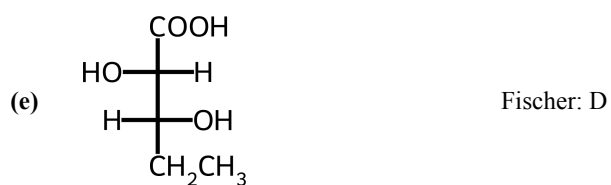
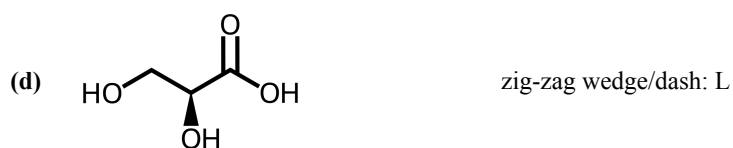


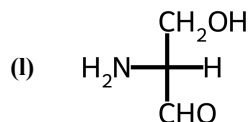
40. Answer: 4

How many of the following compounds, drawn in different projection formulas, have L-configuration? (Convert mentally to a Fischer projection with the most oxidised carbon, $-\text{CHO}$ or $-\text{COOH}$, at the top; the L/L label is decided by the stereocentre farthest from it.)

Method: in a Fischer projection with the most oxidised carbon (CHO or COOH) at the top, look at the stereocentre farthest from the top. $-\text{OH}$ (or $-\text{NH}_2$) on the right means D, on the left means L. Newman, sawhorse and wedge drawings must first be converted to a Fischer projection.





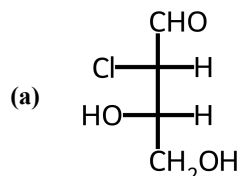


Fischer (rotated 180°): D

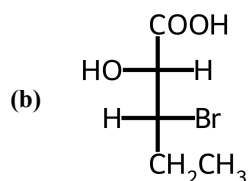
41. Answer: 6

How many of the following compounds, drawn in different projection formulas, are ERYTHRO isomers? (Convert to a Fischer projection: the two non-hydrogen substituents on the adjacent stereocentres lie on the same side for erythro.)

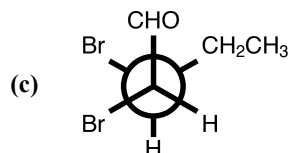
Method: convert each structure to a Fischer projection. If the two similar (non-hydrogen) substituents of the adjacent stereocentres are on the same side the compound is erythro; on opposite sides it is threo.



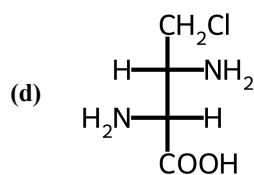
Fischer: erythro



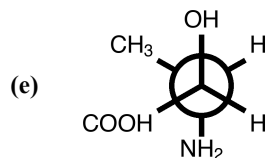
Fischer: threo



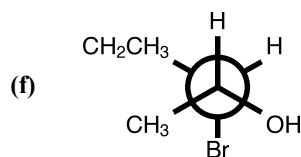
Newman: erythro



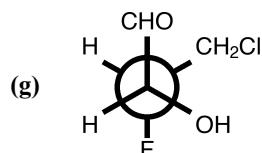
Fischer: threo



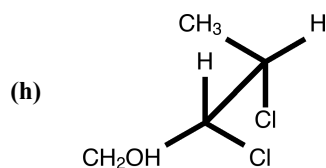
Newman: threo



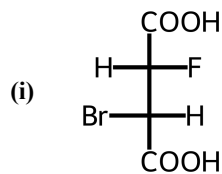
Newman: erythro



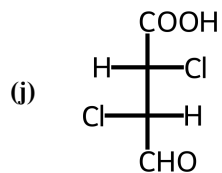
Newman: erythro



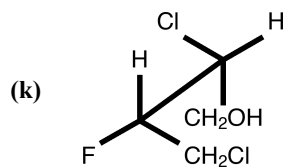
sawhorse: erythro



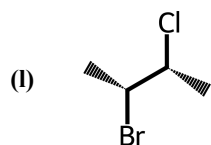
Fischer: threo



Fischer (rotated 180°): threo



sawhorse: erythro

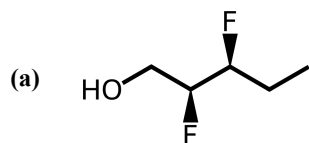


zig-zag wedge/dash: threo

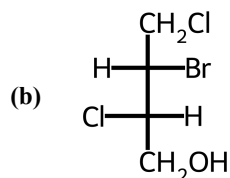
42. Answer: 5

How many of the following compounds, drawn in different projection formulas, are THREO isomers? (Convert to a Fischer projection: the two non-hydrogen substituents on the adjacent stereocentres lie on the opposite side for threo.)

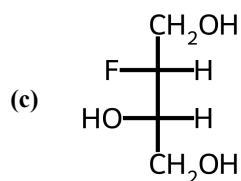
Method: convert each structure to a Fischer projection. If the two similar (non-hydrogen) substituents of the adjacent stereocentres are on the same side the compound is erythro; on opposite sides it is threo.



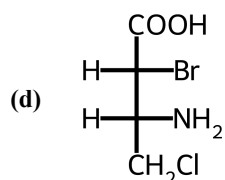
zig-zag wedge/dash: threo



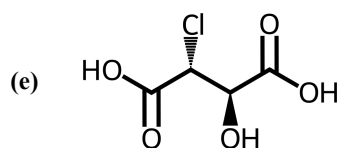
Fischer (rotated 180°): threo



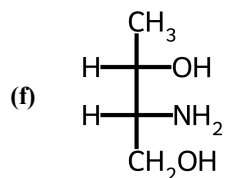
Fischer (rotated 180°): erythro



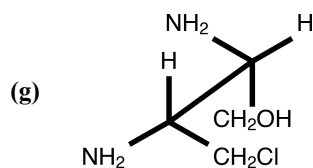
Fischer (rotated 180°): erythro



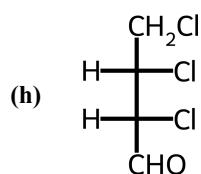
zig-zag wedge/dash: erythro



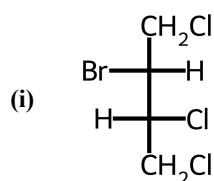
Fischer (rotated 180°): erythro



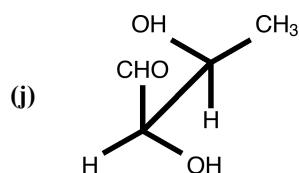
sawhorse: erythro



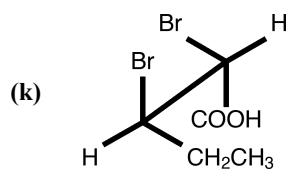
Fischer (rotated 180°): erythro



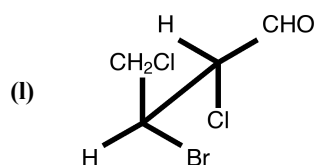
Fischer: threo



sawhorse: threo



sawhorse: threo

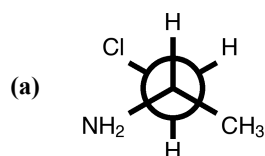


sawhorse: erythro

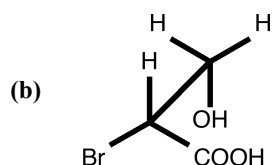
43. Answer: 7

How many of the following compounds, drawn as Newman or sawhorse projections, have 'R' configuration at the stereogenic carbon? (Newman and sawhorse show the stereogenic carbon in front.)

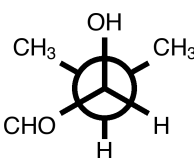
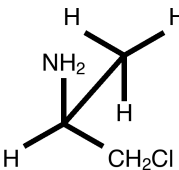
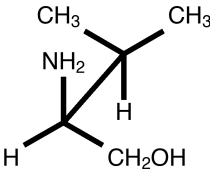
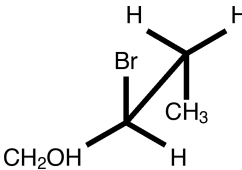
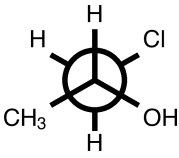
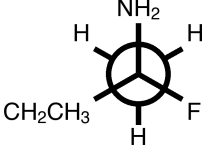
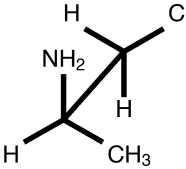
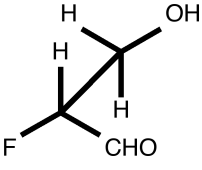
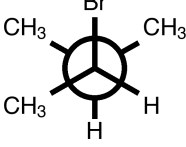
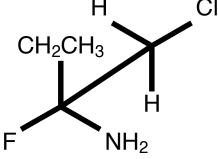
Method (CIP rules): rank the four groups by atomic number of the atom directly attached; on a tie go outwards to the next atoms (duplicate atoms for double bonds). View with the lowest-priority group pointing away; 1→2→3 clockwise is R, anticlockwise is S. For a wedge/dash drawing swap parity if the lowest priority group points towards you.



Newman: S



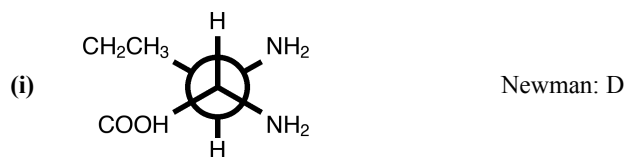
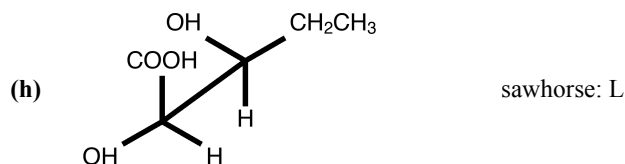
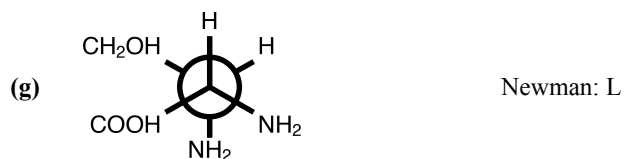
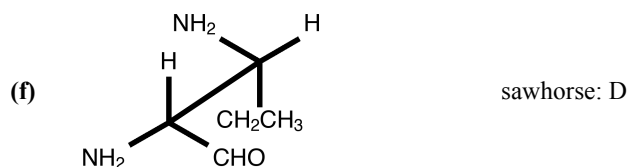
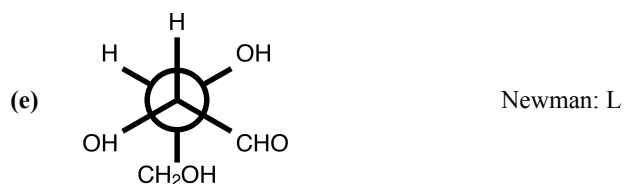
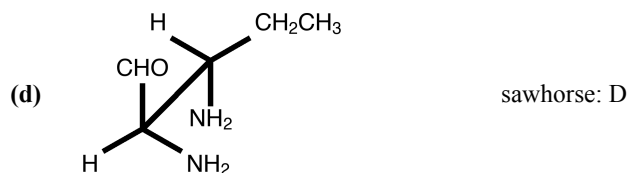
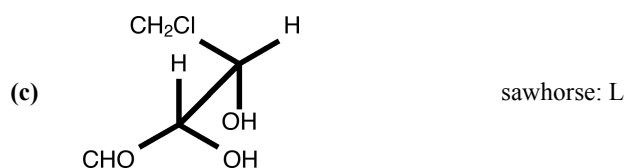
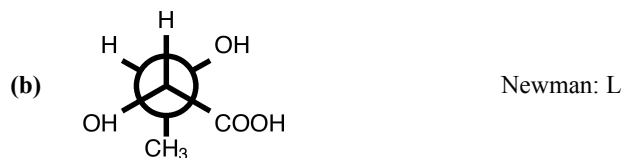
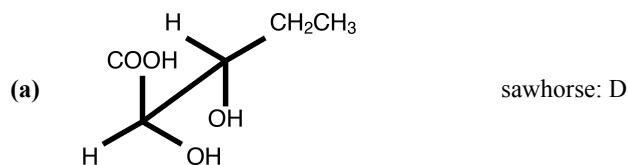
sawhorse: R

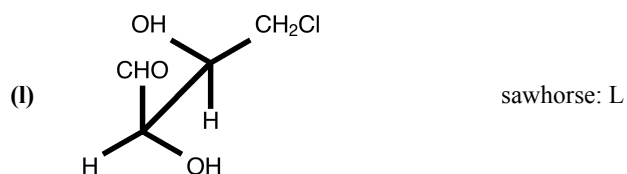
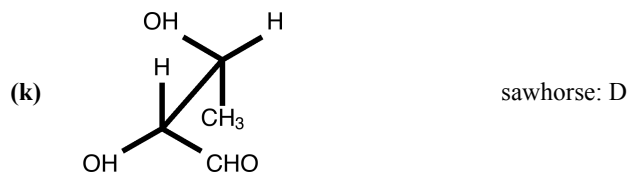
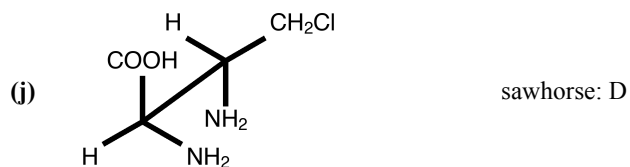
- (c)  Newman: R
- (d)  sawhorse: S
- (e)  sawhorse: S
- (f)  sawhorse: R
- (g)  Newman: R
- (h)  Newman: R
- (i)  sawhorse: R
- (j)  sawhorse: R
- (k)  Newman: S
- (l)  sawhorse: S

44. Answer: 6

How many of the following compounds, drawn as Newman or sawhorse projections, have D-configuration? (Convert mentally to a Fischer projection with the most oxidised carbon, $-\text{CHO}$ or $-\text{COOH}$, at the top; the D/L label is decided by the stereocentre farthest from it.)

Method: in a Fischer projection with the most oxidised carbon (CHO or COOH) at the top, look at the stereocentre farthest from the top. $-\text{OH}$ (or $-\text{NH}_2$) on the right means D, on the left means L. Newman, sawhorse and wedge drawings must first be converted to a Fischer projection.

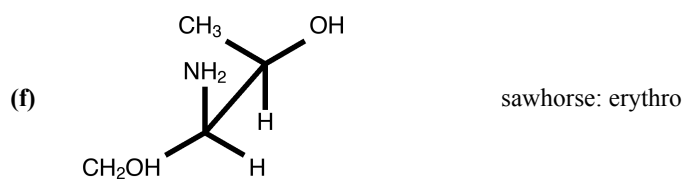
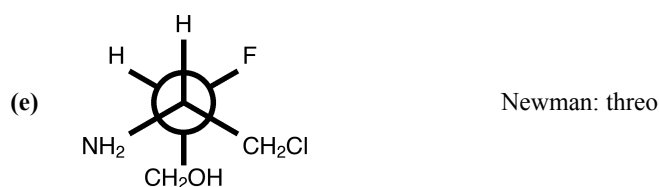
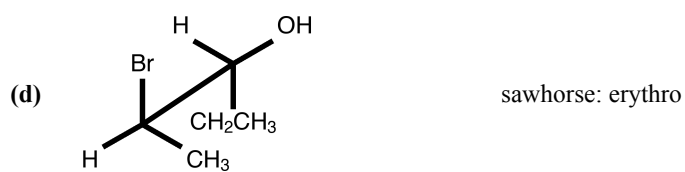
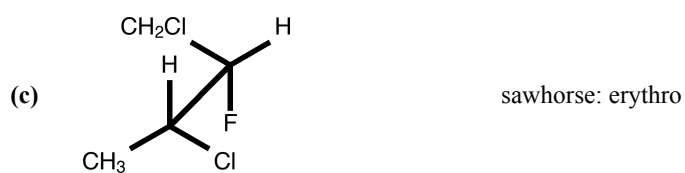
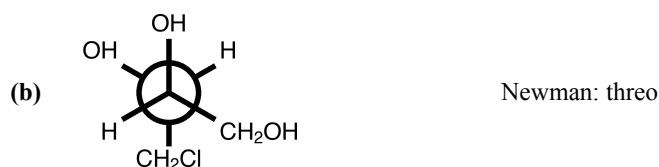
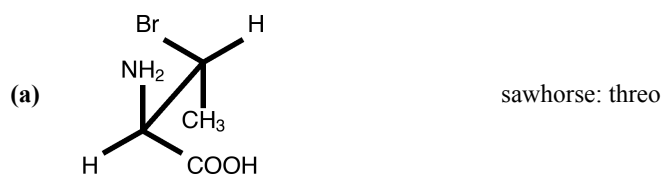


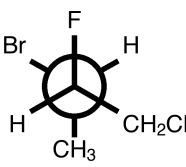
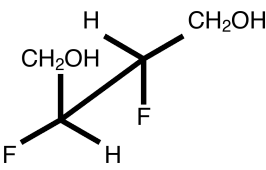
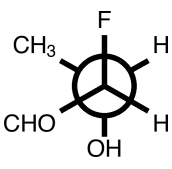
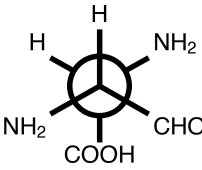
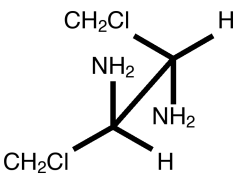
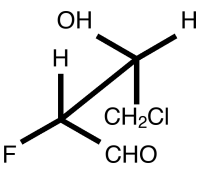


45. Answer: 4

How many of the following compounds, drawn as Newman or sawhorse projections, are ERYTHRO isomers? (Convert to a Fischer projection: the two non-hydrogen substituents on the adjacent stereocentres lie on the same side for erythro.)

Method: convert each structure to a Fischer projection. If the two similar (non-hydrogen) substituents of the adjacent stereocentres are on the same side the compound is erythro; on opposite sides it is threo.



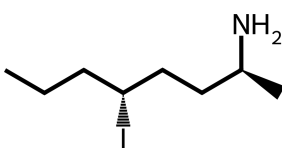
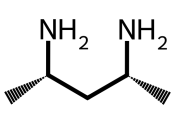
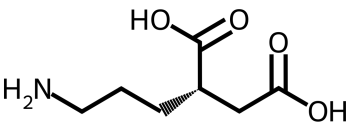
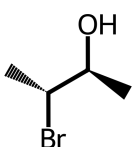
- (g)  Newman: threo
- (h)  sawhorse: threo
- (i)  Newman: threo
- (j)  Newman: threo
- (k)  sawhorse: threo
- (l)  sawhorse: erythro

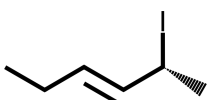
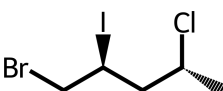
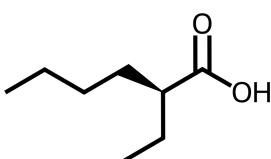
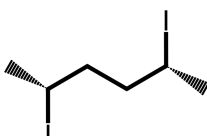
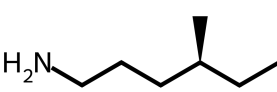
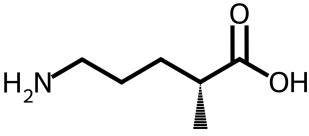
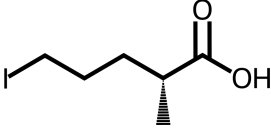
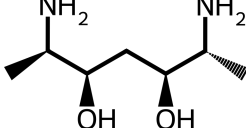
Part E — Chiral centres and substitution products

46. Answer: 20

Find the total number of chiral (asymmetric) carbon centres present in the following acyclic compounds (a chiral centre = carbon bonded to four constitutionally different groups).

Method: an asymmetric molecule has no symmetry element at all except the identity (point group C_1). Test for planes, a centre, proper and alternate axes; count only those with none.

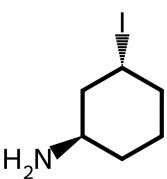
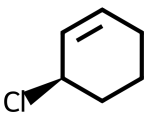
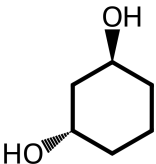
- (a)  2
- (b)  2
- (c)  1
- (d)  2

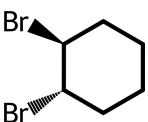
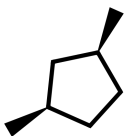
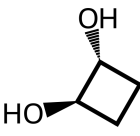
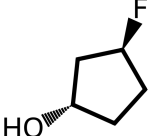
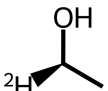
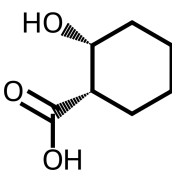
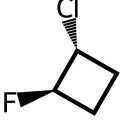
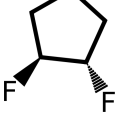
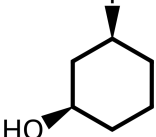
(e)		1
(f)		2
(g)		1
(h)		2
(i)		1
(j)		1
(k)		1
(l)		4

47. Answer: 22

Find the total number of chiral (asymmetric) carbon centres present in all the following compounds (a chiral centre = carbon bonded to four constitutionally different groups).

Method: an asymmetric molecule has no symmetry element at all except the identity (point group C_1). Test for planes, a centre, proper and alternate axes; count only those with none.

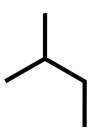
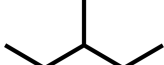
(a)		2
(b)		3-chlorocyclohexene: 1
(c)		2

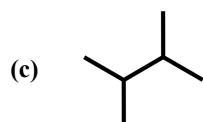
- (d)  1,2-dibromocyclohexane: 2
- (e)  1,3-dimethylcyclopentane: 2
- (f)  2
- (g)  2
- (h)  CH₃CHDOH: 1
- (i)  2
- (j)  2
- (k)  2
- (l)  2

48. Answer: 28

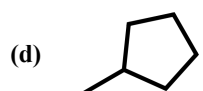
Find the total number of OPTICALLY ACTIVE MONOCHLORO derivatives (structural isomers and stereoisomers, including both enantiomers) obtainable by replacing hydrogen atoms with chlorine, summed over: (a) 2-methylbutane; (b) 3-methylpentane; (c) 2,3-dimethylbutane; (d) methylcyclopentane; (e) 2-methylpentane; (f) butane.

Method: replace H atoms by Cl, collect the distinct structural isomers (use the symmetry of the parent to avoid double counting), then for each isomer count the stereoisomers and keep the optically active ones (both enantiomers of each chiral isomer).

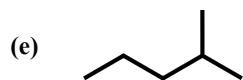
- (a)  2-methylbutane: 4 structural isomers, 6 stereoisomers in all, 4 optically active
- (b)  3-methylpentane: 4 structural isomers, 8 stereoisomers in all, 6 optically active



2,3-dimethylbutane: 2 structural isomers, 3 stereoisomers in all, 2 optically active



methylcyclopentane: 4 structural isomers, 10 stereoisomers in all, 8 optically active



2-methylpentane: 5 structural isomers, 8 stereoisomers in all, 6 optically active



butane: 2 structural isomers, 3 stereoisomers in all, 2 optically active

49. Answer: 14

Find the total number of OPTICALLY ACTIVE DICHLORO derivatives (structural isomers and stereoisomers, including both enantiomers) obtainable by replacing hydrogen atoms with chlorine, summed over: (a) propane; (b) butane; (c) 2-methylpropane; (d) cyclobutane; (e) cyclopentane.

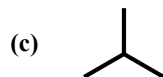
Method: replace H atoms by Cl, collect the distinct structural isomers (use the symmetry of the parent to avoid double counting), then for each isomer count the stereoisomers and keep the optically active ones (both enantiomers of each chiral isomer).



propane: 4 structural isomers, 5 stereoisomers in all, 2 optically active



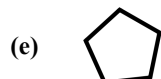
butane: 6 structural isomers, 10 stereoisomers in all, 6 optically active



2-methylpropane: 3 structural isomers, 3 stereoisomers in all, 0 optically active



cyclobutane: 3 structural isomers, 6 stereoisomers in all, 2 optically active



cyclopentane: 3 structural isomers, 7 stereoisomers in all, 4 optically active

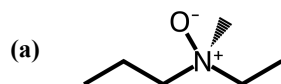
Total optically active = 14

Part F — Stereocentres other than carbon

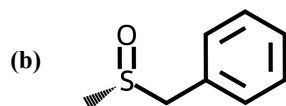
50. Answer: 8

How many of the following species are chiral (optically active as drawn, each stereocentre being one enantiomer)? Treat nitrogen with three substituents + lone pair as freely inverting.

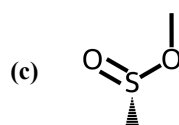
Method: a stereogenic heteroatom needs four different 'groups' (a lone pair counts as one) and a high inversion barrier. P, As and S(IV)/S⁺ centres are stable; neutral amines invert rapidly and cannot be resolved; quaternary N has no lone pair and is stable.



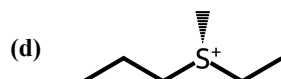
ethylmethylpropylamine N-oxide: amine oxide: N has four different groups and cannot invert → counts



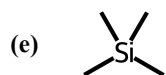
benzyl methyl sulfoxide: S with three different groups and a lone pair → counts



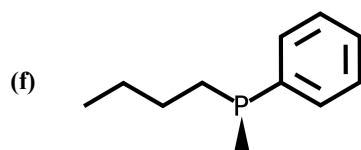
methyl methanesulfonate: S(IV) ester with three different groups + lone pair → counts



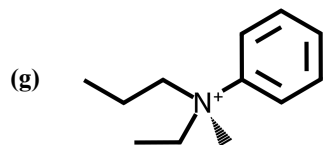
ethylmethylpropylsulfonium ion: sulfonium S: three different groups + lone pair, stable → counts



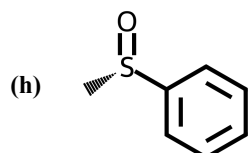
tetramethylsilane: four identical groups → does not count



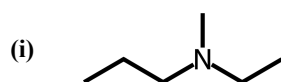
butylmethylphenylphosphine: P with three different groups: slow inversion, resolvable → counts



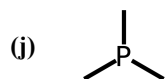
methylpropylphenylanilinium-type ion: quaternary N with four different groups → counts



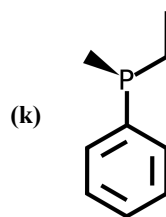
methyl phenyl sulfoxide: S(IV) with O, CH₃, C₆H₅ and a lone pair → counts



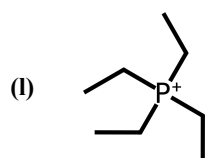
ethylmethylpropylamine: three different groups but N inverts rapidly: not resolvable → does not count



trimethylphosphine: three identical groups → does not count



methylethylphenylphosphine: phosphines invert very slowly: configurationally stable P stereocentre → counts

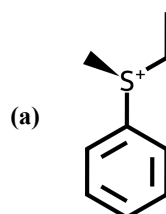


tetraethylphosphonium ion: four identical groups → does not count

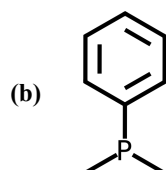
51. Answer: 4

How many of the following can be resolved into stable enantiomers at room temperature?

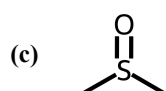
Method: a compound can be resolved if it is chiral and its enantiomers are configurationally stable at room temperature (no fast inversion/rotation). Species with a pyramidal N of three different groups invert rapidly and cannot be resolved.



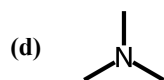
ethyl methyl phenyl sulfonium ion: sulfonium S: three different groups + lone pair → counts



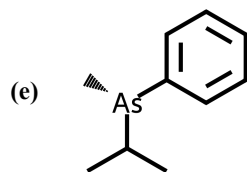
dimethylphenylphosphine: two identical methyl groups on P → does not count



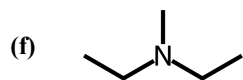
dimethyl sulfoxide: two identical groups on S → does not count



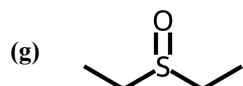
N-chloro... trimethylamine: three identical groups → does not count



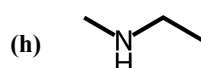
isopropylmethylphenylarsine: As with three different groups: configurationally stable → counts



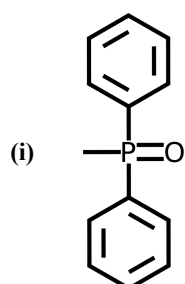
diethylmethanamine: two identical ethyl groups; N inverts rapidly → does not count



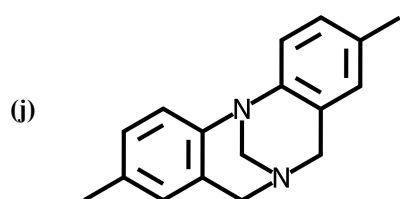
diethyl sulfoxide: two identical ethyl groups on S → does not count



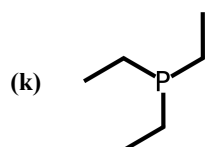
ethylmethanamine: rapid N inversion; no stereocentre → does not count



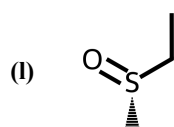
diphenylmethylphosphine oxide: two identical phenyl groups on P → does not count



Tröger's base: bridgehead N atoms cannot invert: both N are fixed stereocentres (C2-chiral cage) → counts



triethylphosphine: three identical groups → does not count



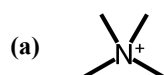
ethyl methyl sulfoxide: S with O, CH₃, C₂H₅ and a lone pair → counts

52. Answer: 4

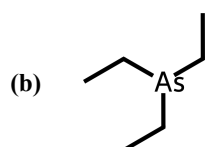
How many of the following are chiral because of a stereogenic atom of group 15 (N, P or As) — not S or Si?

Method: a stereogenic heteroatom needs four different 'groups' (a lone pair counts as one) and a high inversion barrier.

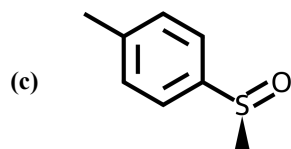
P, As and S(IV)/S⁺ centres are stable; neutral amines invert rapidly and cannot be resolved; quaternary N has no lone pair and is stable.



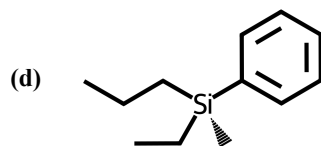
tetramethylammonium ion: four identical groups [stereogenic atom: none] → does not count



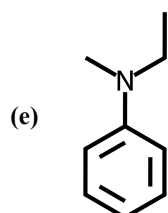
triethylarsine: three identical groups [stereogenic atom: none] → does not count



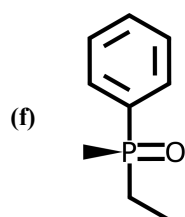
methyl p-tolyl sulfoxide: S in sulfoxide is a stable pyramidal stereocentre (lone pair acts as 4th group) [stereogenic atom: S] → does not count



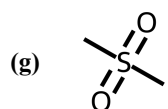
ethylmethylpropylphenylsilane: Si with four different groups behaves like carbon [stereogenic atom: Si] → does not count



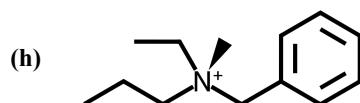
N-ethyl-N-methylaniline: neutral amine: rapid pyramidal inversion (~25 kJ/mol) racemises it [stereogenic atom: none] → does not count



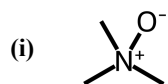
ethylmethylphenylphosphine oxide: P with four different groups [stereogenic atom: P] → counts



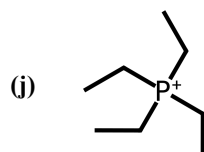
methylphenylsulfoxonium-like: dimethylsulfoxide: two identical groups; two O [stereogenic atom: none] → does not count



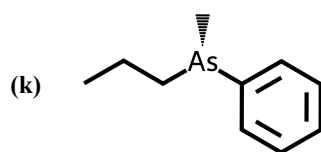
benzyethylmethylpropylammonium ion: quaternary N with four different groups (no lone pair, no inversion) [stereogenic atom: N] → counts



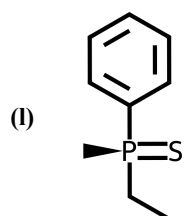
trimethylamine N-oxide: three identical methyl groups [stereogenic atom: none] → does not count



tetraethylphosphonium ion: four identical groups [stereogenic atom: none] → does not count



methylpropylphenylarsine: As stereocentre stable (high inversion barrier) [stereogenic atom: As] → counts

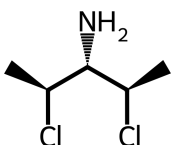
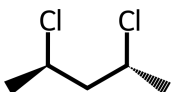
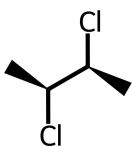
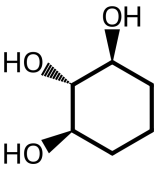
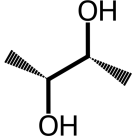
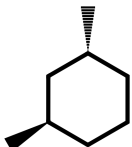
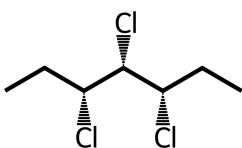
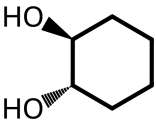
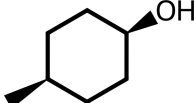
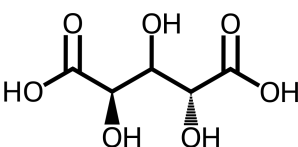
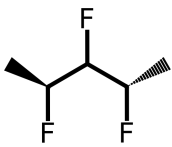


methylethylphenylphosphine sulfide: P with four different groups [stereogenic atom: P] → counts

53. Answer: 5

How many of the following structures contain a PSEUDOASYMMETRIC centre (a stereogenic atom bearing two constitutionally identical but oppositely configured groups, labelled r or s)?

Method: an asymmetric molecule has no symmetry element at all except the identity (point group C1). Test for planes, a centre, proper and alternate axes; count only those with none.

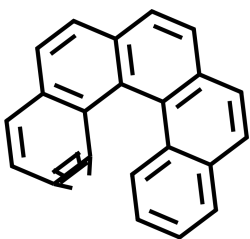
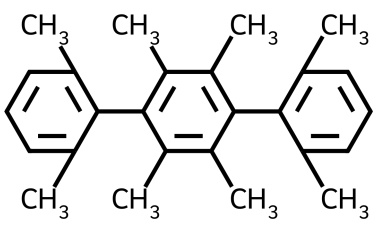
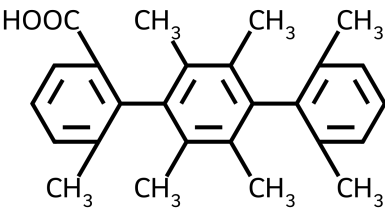
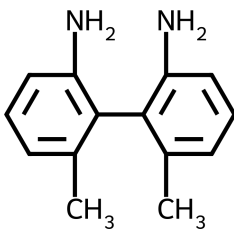
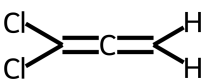
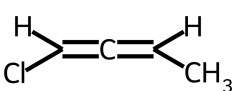
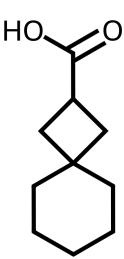
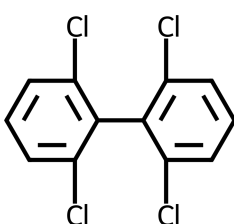
- (a)  2,4-dichloropentan-3-amine: pseudoasymmetric r/s centre present
- (b)  2,4-dichloropentane: none
- (c)  2,3-dichlorobutane: none
- (d)  cyclohexane-1,2,3-triol: pseudoasymmetric r/s centre present
- (e)  2,3-butanediol: none
- (f)  1,3-dimethylcyclohexane: none
- (g)  3,4,5-trichloroheptane: pseudoasymmetric r/s centre present
- (h)  1,2-cyclohexanediol: none
- (i)  4-methylcyclohexanol: pseudoasymmetric r/s centre present
- (j)  2,3,4-trihydroxy-glutaric acid: none
- (k)  2,3,4-trifluoropentane: none
- (l)  1,3-dichlorocyclobutane: pseudoasymmetric r/s centre present

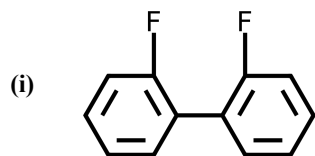
Part G — Spiranes, bicyclics, allenes, biphenyls, terphenyls

54. Answer: 7

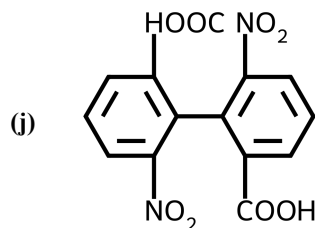
How many of the following can show optical isomerism (some stereoisomer of the skeleton drawn is optically active)?

Method: a compound can show optical isomerism if at least one stereoisomer of the skeleton drawn is chiral. Mark every stereogenic centre (an sp^3 atom with four different groups) and any chiral axis, write the possible stereoisomers and test them for a plane, centre or alternate axis of symmetry. The structures are drawn flat, so all possible stereoisomers must be considered.

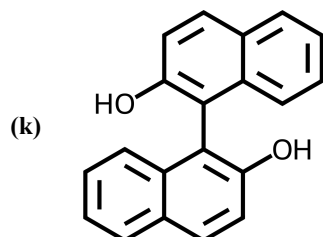
- (a)  hexahelicene: helical, non-planar: P and M enantiomers → counts
- (b)  terphenyl, both ends with identical ortho groups: no stereogenic axis → does not count
- (c)  terphenyl, one end with identical ortho groups: only one axis is stereogenic: 2 enantiomers → counts
- (d)  2,2'-diamino-6,6'-dimethylbiphenyl: two different ortho groups on each ring: stable atropisomers → counts
- (e)  1,1-dichloroallene: identical groups on a terminus: achiral → does not count
- (f)  1-chloro-3-methylallene: each terminus has H and a different group: axial chirality → counts
- (g)  spiro[3.5]nonane-2-carboxylic acid: only one ring substituted: plane of symmetry → does not count
- (h)  2,2',6,6'-tetrachlorobiphenyl: identical ortho groups on each ring: achiral → does not count



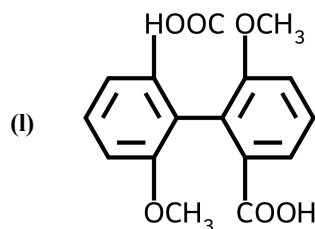
2,2'-difluorobiphenyl: single small ortho groups: free rotation → does not count



6,6'-dinitrodiphenic acid: ortho-tetrasubstituted biphenyl, each ring with two different ortho groups: 2 enantiomers → counts



1,1'-bi-2-naphthol (BINOL): hindered 1,1'-binaphthyl axis: (R)- and (S)-BINOL → counts

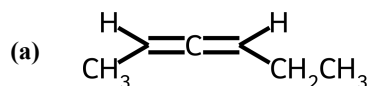


6,6'-dimethoxydiphenic acid: two different ortho groups on each ring: axial chirality → counts

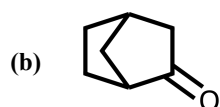
55. Answer: 4

How many of the following can be optically active even though they contain NO stereogenic (asymmetric) carbon atom at all?

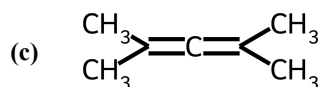
Method: an asymmetric molecule has no symmetry element at all except the identity (point group C_1). Test for planes, a centre, proper and alternate axes; count only those with none.



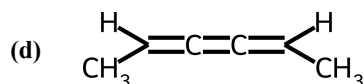
hexa-2,3-diene: unsymmetrical allene, each terminus with two different groups → counts



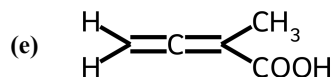
norbornan-2-one: no plane of symmetry; the bridgeheads are linked: one enantiomeric pair → does not count



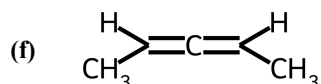
1,1,3,3-tetramethylallene: identical groups on each terminus → does not count



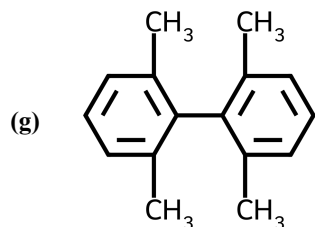
hexa-2,3,4-triene: odd cumulene (3 C=C): planar, cis/trans isomerism only, no optical isomers → does not count



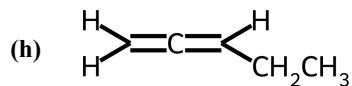
2-methylbuta-2,3-dienoic acid: one terminus carries two H atoms: achiral → does not count



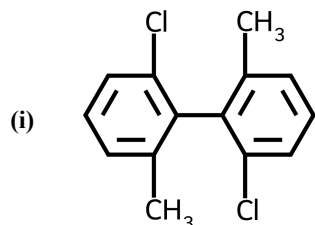
penta-2,3-diene: allene, each terminus H/CH3: axial chirality, 2 enantiomers → counts



2,2',6,6'-tetramethylbiphenyl: identical ortho groups on each ring: achiral (plane through the rings) → does not count



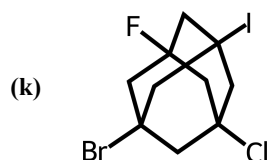
penta-1,2-diene: one terminus carries two H atoms: achiral → does not count



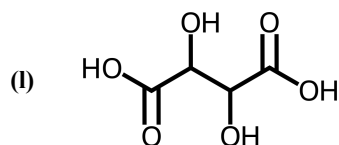
2,2'-dichloro-6,6'-dimethylbiphenyl: restricted rotation, different ortho groups on each ring: 2 enantiomers → counts



norbornane: plane of symmetry, no stereoisomers → does not count



1-bromo-3-chloro-5-fluoro-7-iodoadamantane: four different bridgehead groups on a tetrahedral cage: chiral → counts

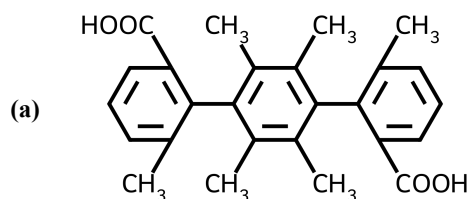


tartaric acid: two stereocentres: (R,R), (S,S) and meso = 3 stereoisomers → does not count

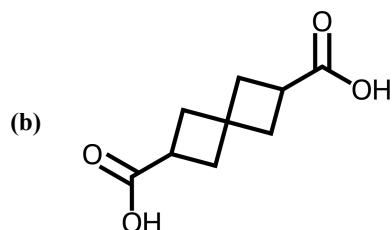
56. Answer: 11

Find the total number of stereoisomers possible for all the compounds (a), (b), (c) and (d) taken together.

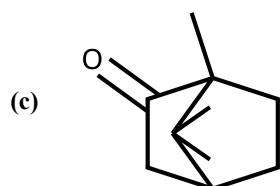
Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.



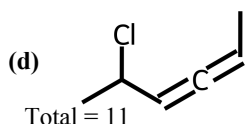
terphenyl, identical ends (each ortho COOH/CH₃): 3 — two stereogenic axes, identical ends: (a,a),(b,b) enantiomers + meso (a,b) = 3 stereoisomers



spiro[3.3]heptane-2,6-dicarboxylic acid: 2 — spiro with a substituent on each ring: axial chirality (2 enantiomers)



camphor: 2 — two bridgehead stereocentres but bridging allows only (+) and (−)

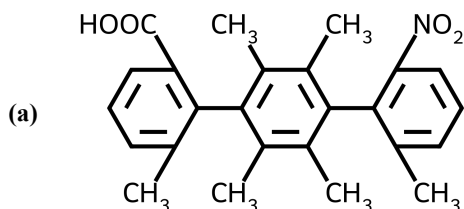


$\text{CH}_3\text{CH}=\text{C}=\text{CH}-\text{CHCl}-\text{CH}_3$: 4 — axial chirality of the allene $\times$ one stereocentre = 4

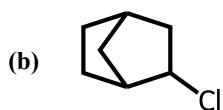
57. Answer: 19

If a, b, c and d are the numbers of stereoisomers of compounds (a), (b), (c) and (d) respectively, find $a + b + c + d$.

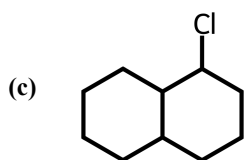
Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.



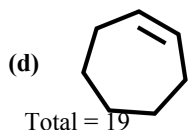
terphenyl, different ends: 4 — two stereogenic axes with different ends: $2^2 = 4$ (two enantiomeric pairs)



2-chloronorbornane: 4 — exo/endo $\times$ two enantiomers = 4 (bridgehead pair is linked)



1-chlorodecalin: 8 — three stereocentres, all combinations possible: 8

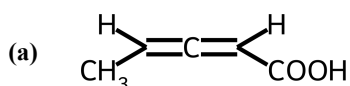


cyclooctene: 3 — cis form + trans form (chiral, resolvable): 3 stereoisomers

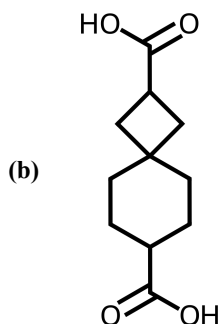
58. Answer: 10

Find the total number of stereoisomers of (a), (b), (c) and (d) taken together.

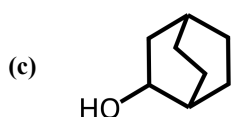
Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.



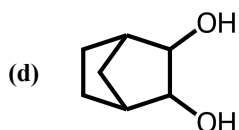
penta-2,3-dienoic acid: 2 — each terminus has H and a different group: axial chirality



spiro[3.5]nonane-2,7-dicarboxylic acid: 2 — a substituent on each ring, para to the spiro atom: axial chirality



bicyclo[2.2.2]octan-2-ol: 2 — C2 is the only stereocentre; the bridgeheads are not stereogenic



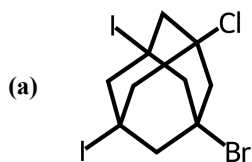
norbornane-2,3-diol: 4 — cis-exo and cis-endo (both meso) + one trans pair

Total = 10

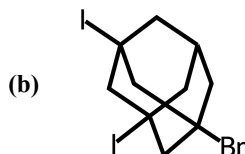
59. Answer: 2

Each adamantane carries groups on its four bridgehead carbons (H where nothing is drawn). How many of the following are optically active?

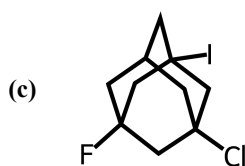
Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.



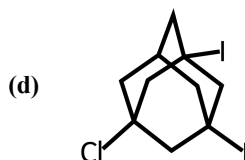
bridgehead groups ['I', 'Br', 'I', 'Cl']: pattern (2, 1, 1) → POS=1, COS=0, AOS=0, AAOS=0



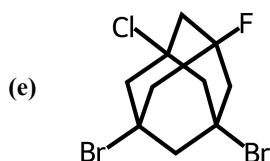
bridgehead groups ['I', 'H', 'I', 'Br']: pattern (2, 1, 1) → POS=1, COS=0, AOS=0, AAOS=0



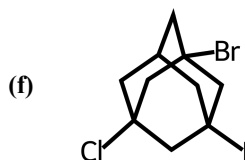
bridgehead groups ['Cl', 'F', 'I', 'H']: pattern (1, 1, 1, 1) → POS=0, COS=0, AOS=0, AAOS=0



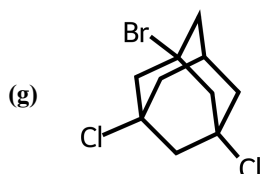
bridgehead groups ['I', 'Cl', 'I', 'H']: pattern (2, 1, 1) → POS=1, COS=0, AOS=0, AAOS=0



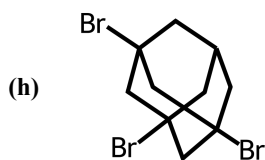
bridgehead groups ['Br', 'Br', 'Cl', 'F']: pattern (2, 1, 1) → POS=1, COS=0, AOS=0, AAOS=0



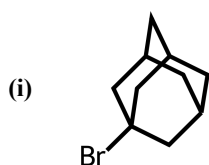
bridgehead groups ['I', 'Cl', 'Br', 'H']: pattern (1, 1, 1, 1) → POS=0, COS=0, AOS=0, AAOS=0



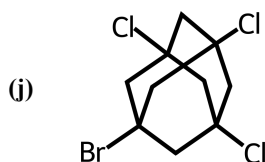
bridgehead groups ['Cl', 'Cl', 'H', 'Br']: pattern (2, 1, 1) → POS=1, COS=0, AOS=0, AAOS=0



bridgehead groups ['Br', 'H', 'Br', 'Br']: pattern (3, 1) → POS=1, COS=0, AOS=1, AAOS=0



bridgehead groups ['H', 'Br', 'H', 'H']: pattern (3, 1) → POS=1, COS=0, AOS=1, AAOS=0

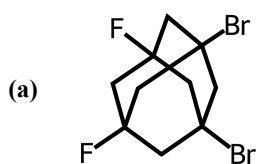


bridgehead groups ['Br', 'Cl', 'Cl', 'Br']: pattern (3, 1) → POS=1, COS=0, AOS=1, AAOS=0

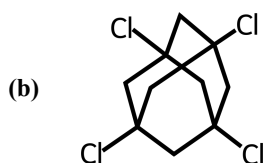
60. Answer: 12

Each adamantane carries groups on its four bridgehead carbons (H where nothing is drawn). Mark POS, COS, AOS and AAOS for each compound and find the total number of marks over all eight compounds.

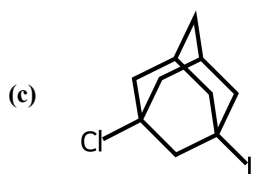
Method: list the symmetry elements of every structure. A plane of symmetry (POS) divides the molecule into two mirror-image halves; a centre of symmetry (COS) maps every atom to an identical atom through the centre; an axis of symmetry (AOS) is a proper rotation axis C_n ($n \geq 2$) after which the molecule looks unchanged; an alternate axis (AAOS) is a rotation-reflection axis S_n ($n \geq 3$). Rings are taken as planar with wedge groups above and dashed groups below the ring plane; for each structure decide which of the four elements exist, then add up the marks.



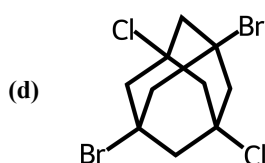
bridgehead groups ['F', 'Br', 'F', 'Br']: pattern (2, 2) → POS=1, COS=0, AOS=1, AAOS=0



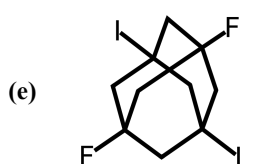
bridgehead groups ['Cl', 'Cl', 'Cl', 'Cl']: pattern (4,) → POS=1, COS=0, AOS=1, AAOS=1



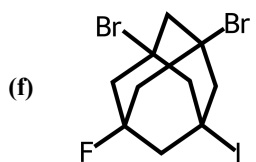
bridgehead groups ['H', 'H', 'I', 'Cl']: pattern (2, 1, 1) → POS=1, COS=0, AOS=0, AAOS=0



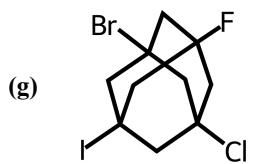
bridgehead groups ['Br', 'Cl', 'Cl', 'Br']: pattern (2, 2) → POS=1, COS=0, AOS=1, AAOS=0



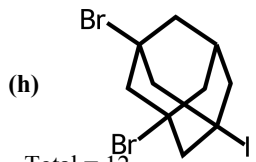
bridgehead groups ['F', 'I', 'I', 'F']: pattern (2, 2) → POS=1, COS=0, AOS=1, AAOS=0



bridgehead groups ['F', 'I', 'Br', 'Br']: pattern (2, 1, 1) → POS=1, COS=0, AOS=0, AAOS=0



bridgehead groups ['I', 'Cl', 'Br', 'F']: pattern (1, 1, 1, 1) → POS=0, COS=0, AOS=0, AAOS=0



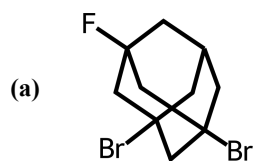
bridgehead groups ['Br', 'H', 'Br', 'I']: pattern (2, 1, 1) → POS=1, COS=0, AOS=0, AAOS=0

Total = 12

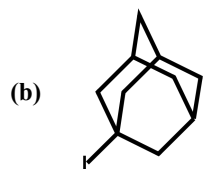
61. Answer: 5

Each adamantane carries groups on its four bridgehead carbons (H where nothing is drawn). How many of the following possess a proper axis of symmetry (C_n , $n \geq 2$)?

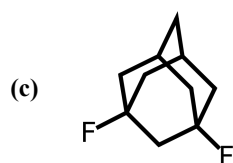
Method: a proper axis C_n rotates the molecule by $360^\circ/n$ into an identical orientation. Check for such rotations (in planar-ring drawings the ring normal and the in-plane axes through opposite atoms or bond mid-points must map every wedge/dash group onto an identical one) and count or add as asked.



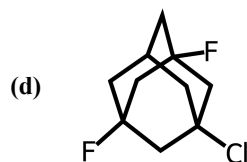
bridgehead groups ['F', 'H', 'Br', 'Br']: pattern (2, 1, 1) $\rightarrow$ POS=1, COS=0, AOS=0, AAOS=0



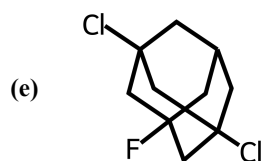
bridgehead groups ['H', 'H', 'I', 'H']: pattern (3, 1) $\rightarrow$ POS=1, COS=0, AOS=1, AAOS=0



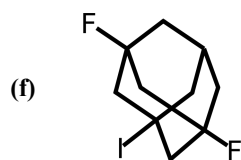
bridgehead groups ['H', 'F', 'F', 'H']: pattern (2, 2) $\rightarrow$ POS=1, COS=0, AOS=1, AAOS=0



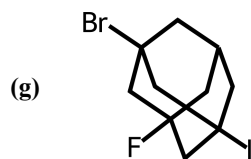
bridgehead groups ['Cl', 'F', 'F', 'H']: pattern (2, 1, 1) $\rightarrow$ POS=1, COS=0, AOS=0, AAOS=0



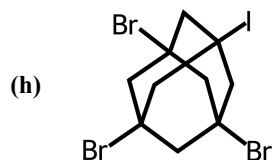
bridgehead groups ['Cl', 'H', 'F', 'Cl']: pattern (2, 1, 1) $\rightarrow$ POS=1, COS=0, AOS=0, AAOS=0



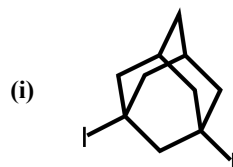
bridgehead groups ['F', 'H', 'I', 'F']: pattern (2, 1, 1) $\rightarrow$ POS=1, COS=0, AOS=0, AAOS=0



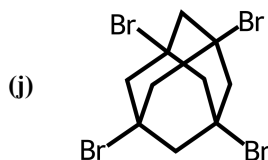
bridgehead groups ['Br', 'H', 'F', 'I']: pattern (1, 1, 1, 1) $\rightarrow$ POS=0, COS=0, AOS=0, AAOS=0



bridgehead groups ['Br', 'Br', 'Br', 'I']: pattern (3, 1) $\rightarrow$ POS=1, COS=0, AOS=1, AAOS=0



bridgehead groups ['I', 'H', 'H', 'I']: pattern (2, 2) $\rightarrow$ POS=1, COS=0, AOS=1, AAOS=0



bridgehead groups ['Br', 'Br', 'Br', 'Br']: pattern (4,) → POS=1, COS=0, AOS=1, AAOS=1

Part H — Counting stereoisomers

62. Answer: 9

How many stereoisomers are possible for the following compound?

Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.

$$x = 9$$

63. Answer: 8

How many stereoisomers are possible for the following compound?

Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.

$$x = 8$$

64. Answer: 32

How many stereoisomers are possible for the following compound?

Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.

$$x = 32$$

65. Answer: 4

The total number of stereoisomers of the compound shown is 2^n . Find n.

Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.

$$x = 16 = 2^4$$

66. Answer: 27

If the total number of stereoisomers of the compound shown is X, find $(X - 5)$.

Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.

$$X = 32$$

67. Answer: 20

For the compound shown let x = total stereoisomers, y = optically active stereoisomers and z = optically inactive stereoisomers. Find $x + y + z$.

Method: a mixture is optically inactive when the rotations cancel: equimolar enantiomers (racemic), or a meso compound, or mixtures of all stereoisomers in equal amounts. An excess of one enantiomer or a mixture of different compounds does not cancel.

$$x=10, y=8, z=2$$

68. Answer: 16

Find the number of pairs of enantiomers possible for the compound shown.

Method: convert both members to the same representation and compare configuration at every stereocentre: all centres identical → same compound (homomers); all centres opposite → enantiomers; some same and some opposite → diastereomers; different connectivity → constitutional isomers. Then apply the property asked (enantiomers share all physical properties except the sign of rotation; diastereomers and constitutional isomers differ).

32 chiral stereoisomers → 16 pairs

69. Answer: 2

Find the number of optically INACTIVE stereoisomers of the compound shown.

Method: a mixture is optically inactive when the rotations cancel: equimolar enantiomers (racemic), or a meso compound, or mixtures of all stereoisomers in equal amounts. An excess of one enantiomer or a mixture of different compounds does not cancel.

$$x=10, \text{ chiral}=8, \text{ inactive}=2$$

70. Answer: 18

If a, b and c are the numbers of stereoisomers of the compounds (a), (b) and (c) shown, find $a + b + c$.

Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.

borneol skeleton (7,7-dimethylnorbornan-2-ol): 4

5-norbornen-2-ol: 4

1,4-dichlorodecalin: 10

Total = 18

71. Answer: 6

Find the total number of optically active stereoisomers of the compounds (a) 2,6-dichloronorbornane and (b) 7,7-dimethylnorbornan-2-ol taken together.

Method: replace H atoms by Cl, collect the distinct structural isomers (use the symmetry of the parent to avoid double counting), then for each isomer count the stereoisomers and keep the optically active ones (both enantiomers of each chiral isomer).

2,6-dichloronorbornane: 2 optically active of 4

borneol skeleton (7,7-dimethylnorbornan-2-ol): 4 optically active of 4

72. Answer: 104

For the compound shown find (number of diastereomeric pairs) – (number of enantiomeric pairs) among all its stereoisomers (each unordered pair counted once).

Method: enumerate all stereoisomers, find the chiral ones (optically active; they form pairs of enantiomers), the achiral ones (meso/inactive) and note that all other pairs are diastereomeric: total pairs = $x(x-1)/2$.

$$x=16, \text{ all pairs}=120, \text{ enantiomeric pairs}=8, \text{ diastereomeric pairs}=112$$

73. Answer: 8

Find the number of optically active stereoisomers of the compound shown.

Method: enumerate all stereoisomers, find the chiral ones (optically active; they form pairs of enantiomers), the achiral ones (meso/inactive) and note that all other pairs are diastereomeric: total pairs = $x(x-1)/2$.

$$x=8, \text{ optically active}=8$$

74. Answer: 48

Calculate $x + y + z$ where x = total stereoisomers, y = optically inactive stereoisomers and z = pairs of enantiomers, for the compound shown.

Method: convert both members to the same representation and compare configuration at every stereocentre: all centres identical → same compound (homomers); all centres opposite → enantiomers; some same and some opposite → diastereomers; different connectivity → constitutional isomers. Then apply the property asked (enantiomers share all physical properties except the sign of rotation; diastereomers and constitutional isomers differ).

$$x=32, y=0, z=16$$

Part I — Optical rotation and optical purity

75. Answer: 13.3

The specific rotation of sucrose is $+66.5^\circ$. What is the observed rotation (in degrees) of a solution containing 10 g of sucrose per 100 mL in a 20 cm polarimeter tube?

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \sum(\text{mole fraction} \times [\alpha])$.

$$\alpha = [\alpha] \cdot l \cdot c = 66.5 \times 2 \text{ dm} \times 0.10 \text{ g/mL} = 13.3^\circ$$

76. Answer: 48

A solution of 5 g of an optically active compound in 100 mL of solvent, placed in a 10 cm tube, gives an observed rotation of $+2.4^\circ$. Find its specific rotation (in degrees).

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$[\alpha] = \alpha/(l \cdot c) = 2.4/(1 \times 0.05) = 48^\circ$$

77. Answer: 36

The specific rotation of pure (+)-A is $+60^\circ$. Find the specific rotation of a mixture containing 80% (+)-A and 20% (–)-A.

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$[\alpha]_{\text{mix}} = 60 \times (0.80 - 0.20) = 36^\circ$$

78. Answer: 8

3 g of (+)-X and 1 g of (–)-X are dissolved together in a total volume of 20 mL and placed in a 20 cm tube. The specific rotation of pure (+)-X is $+40^\circ$. Find the observed rotation (in degrees).

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$\alpha = 40 \times 2 \times (3/20) - 40 \times 2 \times (1/20) = 12 - 4 = 8^\circ$$

79. Answer: -52

A compound has specific rotation -52° . What is the observed rotation (in degrees) for a 0.4 g/mL solution in a 25 cm tube?

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$\alpha = -52 \times 2.5 \text{ dm} \times 0.4 = -52^\circ$$

80. Answer: 80

A partially racemised sample has $[\alpha] = +18^\circ$, while the pure (+)-enantiomer has $[\alpha] = +30^\circ$. What percentage of the sample is the (+)-enantiomer?

Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.

$$\text{Optical purity} = 18/30 \times 100 = 60\% = \text{ee. } \%(+) = (100 + \text{ee})/2 = 80\%$$

81. Answer: 2

How many grams of (–)-enantiomer must be mixed with 6 g of the (+)-enantiomer to obtain a mixture of 50% optical purity?

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$\text{ee} = (6 - x)/(6 + x) = 0.50 \rightarrow x = 2 \text{ g}$$

82. Answer: 52.48

α -D-glucose has $[\alpha] = +112^\circ$ and β -D-glucose has $[\alpha] = +19^\circ$. At mutarotation equilibrium the mixture contains 36% α and 64% β . Find the equilibrium specific rotation (in degrees).

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$[\alpha]_{\text{eq}} = 0.36 \times 112 + 0.64 \times 19 = 52.48^\circ$$

83. Answer: 10

A solution of 0.5 g/mL in a 10 cm tube rotates plane-polarised light by $+10^\circ$. What will be the rotation (in degrees) if the concentration is changed to 0.2 g/mL and a 25 cm tube is used?

Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.

$$\alpha \text{ varies as } l \cdot c \rightarrow \alpha = 10 \times (25/10) \times (0.2/0.5) = 10^\circ$$

84. Answer: 13.5

6 g of 2-butanol are dissolved in a total volume of 40 mL and filled in a 200 mm tube. The observed rotation is +4.05°. Find the specific rotation of 2-butanol (in degrees).

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$c = 6/40 = 0.15 \text{ g/mL}, l = 2 \text{ dm}; [\alpha] = 4.05/(2 \times 0.15) = 13.5^\circ$$

85. Answer: 25

A mixture of enantiomers shows $[\alpha] = +9^\circ$ while the pure enantiomer has $[\alpha] = +36^\circ$. Find its optical purity (%).

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$\text{OP} = 9/36 \times 100 = 25\%$$

86. Answer: 30

A sample has 40% optical purity and the pure enantiomer has $[\alpha] = +75^\circ$. What is the observed specific rotation (in degrees)?

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$[\alpha]_{\text{obs}} = 0.40 \times 75 = 30^\circ$$

87. Answer: 50

A sample of 2-butanol has $[\alpha] = -6.75^\circ$, whereas pure (–)-2-butanol has $[\alpha] = -13.5^\circ$. What percentage of the sample is racemic (equal amounts of both enantiomers)?

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$\text{OP} = 50\%; \text{racemic part} = 100 - 50 = 50\%$$

88. Answer: 80

A mixture contains 9 mol of (R)- and 1 mol of (S)-enantiomer. Find the enantiomeric excess (%).

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

$$\text{ee} = (9 - 1)/(9 + 1) \times 100 = 80\%$$

Part J — Stereoisomer calculations**89. Answer: 28**

Find the total number of open-chain stereoisomers of (A) an aldotetrose, (B) an aldopentose and (C) an aldohexose (all stereocentres unlike), i.e. $a + b + c$.

Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.

aldotetrose: $4 (2^n)$

aldopentose: $8 (2^n)$

aldohexose: $16 (2^n)$

Total = 28

90. Answer: 16

The compound shown (3,4-diphenylcyclobutane-1,2-dicarboxylic acid) exists in two or more stereoisomeric forms. Let a = total number of stereoisomers, b = number of pairs of enantiomers and c = number of meso compounds. Find $a + b + c$.

Method: a meso compound has two or more stereocentres but is achiral because of an internal plane or centre of symmetry. Look for stereocentres arranged symmetrically with opposite configurations.

$$a = 10, b = 4, c = 2 \text{ (every achiral stereoisomer is meso here)} \rightarrow 16$$

91. Answer: 8

Find the total number of stereoisomers of the biaryl shown (hindered rotation about the aryl–aryl bond, one stereogenic carbon and one double bond).

Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.

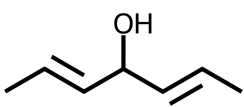
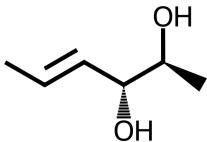
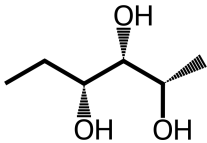
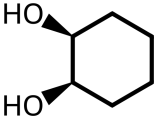
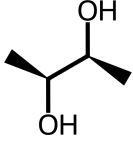
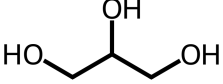
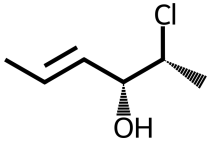
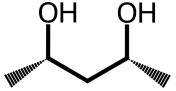
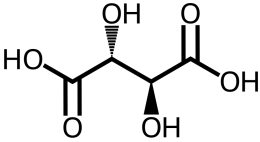
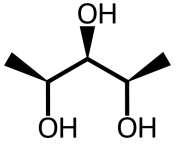
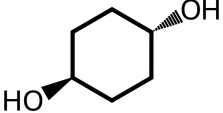
Stereogenic axis (2) $\times$ C=C (E/Z, 2) $\times$ CH(OH) centre (2) = 8.

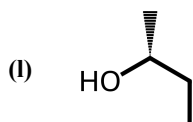
Part K — Resolution and mixtures

92. Answer: 7

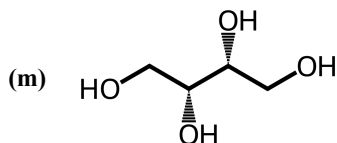
Each alcohol drawn below is a single pure stereoisomer. How many of them give a mixture of DIASTEREOMERIC esters when treated with ($\pm$)-2-methylbutanoic acid?

Method: a racemic acid forms diastereomeric esters only with a CHIRAL alcohol; an achiral or meso alcohol gives two enantiomeric esters. Decide whether each drawn alcohol is chiral.

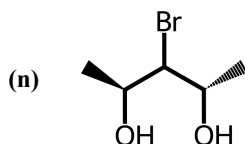
- | | |
|---|---|
| (a)  | achiral/meso alcohol $\rightarrow$ enantiomeric esters $\rightarrow$ does not count |
| (b)  | chiral alcohol $\rightarrow$ diastereomeric esters $\rightarrow$ counts |
| (c)  | chiral alcohol $\rightarrow$ diastereomeric esters $\rightarrow$ counts |
| (d)  | achiral/meso alcohol $\rightarrow$ enantiomeric esters $\rightarrow$ does not count |
| (e)  | chiral alcohol $\rightarrow$ diastereomeric esters $\rightarrow$ counts |
| (f)  | achiral/meso alcohol $\rightarrow$ enantiomeric esters $\rightarrow$ does not count |
| (g)  | chiral alcohol $\rightarrow$ diastereomeric esters $\rightarrow$ counts |
| (h)  | achiral/meso alcohol $\rightarrow$ enantiomeric esters $\rightarrow$ does not count |
| (i)  | achiral/meso alcohol $\rightarrow$ enantiomeric esters $\rightarrow$ does not count |
| (j)  | achiral/meso alcohol $\rightarrow$ enantiomeric esters $\rightarrow$ does not count |
| (k)  | achiral/meso alcohol $\rightarrow$ enantiomeric esters $\rightarrow$ does not count |



chiral alcohol → diastereomeric esters → counts



chiral alcohol → diastereomeric esters → counts



chiral alcohol → diastereomeric esters → counts

93. Answer: 13

Separate equimolar mixtures containing ALL the stereoisomers of (a) 2,3,4,5-tetrachlorohexane, (b) pentane-2,3,4-triol, (c) 2,3-dibromopentane and (d) 3,4-dichlorohexane are each fractionally distilled. Find the total number of fractions obtained for the four mixtures.

Method: enantiomers have the same boiling point and are not separated by distillation, whereas diastereomers and constitutional isomers are. So each racemic pair and each achiral stereoisomer forms one fraction. Chiral stereoisomers are optically active and obtained by a chiral column from the racemic fractions.

2,3,4,5-tetrachlorohexane: 6 fractions (one per enantiomeric pair, one per achiral stereoisomer)

pentane-2,3,4-triol: 3 fractions (one per enantiomeric pair, one per achiral stereoisomer)

2,3-dibromopentane: 2 fractions (one per enantiomeric pair, one per achiral stereoisomer)

3,4-dichlorohexane: 2 fractions (one per enantiomeric pair, one per achiral stereoisomer)

Total = 13

94. Answer: 16

Each fraction of the mixtures in the previous question is passed through a chiral column. How many DIFFERENT optically active compounds are finally isolated from the four mixtures together?

Method: evaluate the property asked for every structure separately (shown below) and then count or add as the question requires.

2,3,4,5-tetrachlorohexane: 8 optically active stereoisomers

pentane-2,3,4-triol: 2 optically active stereoisomers

2,3-dibromopentane: 4 optically active stereoisomers

3,4-dichlorohexane: 2 optically active stereoisomers

Total = 16

95. Answer: 6

Read the following 12 descriptions of mixtures and find in how many of them the mixture is optically INACTIVE.

Method: a mixture is optically inactive when the rotations cancel: equimolar enantiomers (racemic), or a meso compound, or mixtures of all stereoisomers in equal amounts. An excess of one enantiomer or a mixture of different compounds does not cancel.

(1) True — optically inactive: rotations cancel

(2) True — optically inactive: rotations cancel

(3) False — optically active: the rotations do not cancel

(4) False — optically active: the rotations do not cancel

(5) True — optically inactive: rotations cancel

(6) True — optically inactive: rotations cancel

(7) False — optically active: the rotations do not cancel

(8) False — optically active: the rotations do not cancel

(9) True — optically inactive: rotations cancel

(10) False — optically active: the rotations do not cancel

(11) False — optically active: the rotations do not cancel

(12) True — optically inactive: rotations cancel

96. Answer: 10

(2R,3R)-A has $[\alpha] = +30^\circ$, its enantiomer (2S,3S)-A has -30° and the meso diastereomer has 0° . A mixture contains 3 mol (R,R), 1 mol (S,S) and 2 mol meso isomer (all of equal molar mass). Find the specific rotation of the mixture (in degrees).

Method: a meso compound has two or more stereocentres but is achiral because of an internal plane or centre of symmetry. Look for stereocentres arranged symmetrically with opposite configurations.

$$(3 \times 30 + 1 \times (-30) + 2 \times 0) / 6 = 10^\circ$$

97. Answer: -4.5

A compound has two optically active enantiomeric forms with $[\alpha] = \pm 45^\circ$ and a meso form. A sample is prepared by mixing 2 mol of (+), 3 mol of (−) and 5 mol of the meso form. Find the specific rotation (in degrees) of the sample (molar masses are equal).

Method: a meso compound has two or more stereocentres but is achiral because of an internal plane or centre of symmetry. Look for stereocentres arranged symmetrically with opposite configurations.

$$(2 \times 45 - 3 \times 45 + 0) / 10 = -4.5^\circ$$

98. Answer: 18

Three-component mixture in a 10 cm tube: (+)-A ($[\alpha] = +60^\circ$) at 0.5 g/mL and (−)-A at 0.2 g/mL (same compound, so the (−)-form has $[\alpha] = -60^\circ$). Find the observed rotation (in degrees).

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}} / [\alpha]_{\text{pure}}) \times 100 = \text{enantiomeric excess} = \%_{\text{major}} - \%_{\text{minor}}$, and for mixtures $[\alpha]_{\text{mix}} = \sum(\text{mole fraction} \times [\alpha])$.

$$\alpha = 60 \times 1 \times 0.5 - 60 \times 1 \times 0.2 = 18^\circ$$

Part L — Statements on optical isomerism**99. Answer: 8**

Read the following 14 statements on optical activity and chirality and find how many of them are correct.

Method: judge every statement independently using the definitions of chirality, symmetry elements, configuration and properties of stereoisomers; the explanation given with each statement states why it is true or false.

- (1) True — A chiral molecule has a non-superimposable mirror image and rotates plane-polarised light.
- (2) True — One stereocentre alone gives no internal symmetry element.
- (3) False — It may be meso (e.g. meso-tartaric acid).
- (4) True — $\sigma = S_1$: any improper axis means achiral.
- (5) True — $i = S_2$.
- (6) False — (R,R)-tartaric acid has a C_2 axis and is chiral; only improper axes exclude chirality.
- (7) True — S_n includes σ ($n = 1$) and i ($n = 2$).
- (8) False — Allenes, ortho-substituted biphenyls and spiro compounds can be chiral without a stereocentre.
- (9) False — Same reason as above.
- (10) True — Two halves rotate equally and oppositely.
- (11) False — Specific rotation depends on solvent, temperature, wavelength and concentration.
- (12) True — Definition.
- (13) True — Equal and opposite rotations of two separate enantiomers.
- (14) False — Sign of rotation is experimental; R/S is a CIP descriptor and the two are unrelated.

100. Answer: 10

Read the following 14 statements on symmetry, meso compounds and ring stereochemistry and find how many of them are correct.

Method: a meso compound has two or more stereocentres but is achiral because of an internal plane or centre of symmetry. Look for stereocentres arranged symmetrically with opposite configurations.

- (1) True — The two halves are mirror images.
- (2) True — Meso = achiral diastereomer containing stereocentres.
- (3) True — Plane of symmetry between C1 and C2.
- (4) True — No plane or centre of symmetry.
- (5) True — Plane through C2 and C5.
- (6) False — It has a plane (and a centre) of symmetry.
- (7) True — One meso form and one pair of enantiomers.
- (8) False — (2R,3S) is the meso form.
- (9) False — It has three (two enantiomers and one meso).
- (10) True — No symmetry reduction when all centres differ.
- (11) True — Four unlike stereocentres.
- (12) True — The two Br atoms and the two CH₃ groups are related by inversion.

(13) False — They are (R) and (S).

(14) True — Meso compounds are examples.

101. Answer: 9

Read the following 14 statements on CIP rules, Fischer projections and D/L and find how many of them are correct.

Method: judge every statement independently using the definitions of chirality, symmetry elements, configuration and properties of stereoisomers; the explanation given with each statement states why it is true or false.

- (1) True — Go outward only on ties.
- (2) False — -COOH (O,O,O) outranks $\text{-CH}_2\text{OH}$ (O,H,H).
- (3) True — Duplicated atoms on the double bond make vinyl rank higher.
- (4) True — Swapping two groups inverts the configuration.
- (5) False — No fixed relation between R/S and the sign of rotation.
- (6) True — Only 180° in-plane rotations preserve it.
- (7) True — Allowed operation.
- (8) False — D/L is relative to glyceraldehyde and independent of R/S.
- (9) True — Standard assignment.
- (10) True — L-amino acids (except cysteine) are S.
- (11) False — They lie on the same side.
- (12) True — (2R,3S,4R,5R).
- (13) True — Greater mass number breaks the tie.
- (14) False — Cl ($Z = 17$) outranks S ($Z = 16$).

102. Answer: 8

Read the following 14 statements on enantiomers, diastereomers and physical properties and find how many of them are correct.

Method: judge every statement independently using the definitions of chirality, symmetry elements, configuration and properties of stereoisomers; the explanation given with each statement states why it is true or false.

- (1) True — Same scalar properties.
- (2) True — Not mirror images.
- (3) True — The transition states are diastereomeric.
- (4) False — Same boiling point.
- (5) True — Stereoisomers that are not mirror images.
- (6) False — They are diastereomers.
- (7) True — Definition.
- (8) True — 2^n stereoisomers form $2^n - 1$ pairs.
- (9) True — E.g. D-glucose and D-mannose.
- (10) False — Anomers (α and β glucose) are diastereomers.
- (11) False — Barriers are too low.
- (12) True — Definition.
- (13) False — Equal angle, opposite direction.
- (14) False — The racemate can melt differently (racemic compound/conglomerate).

103. Answer: 9

Read the following 14 statements on specific rotation, racemisation and resolution and find how many of them are correct.

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \sum(\text{mole fraction} \times [\alpha])$.

- (1) True — Definition.
- (2) True — $\alpha = [\alpha] \cdot l \cdot c$.
- (3) True — E.g. through diastereomeric salts.
- (4) False — It is achiral.
- (5) True — Equals % enantiomeric excess.
- (6) True — $75 - 25 = 50$.
- (7) True — ee = 50% $\rightarrow$ 75:25.
- (8) False — Rotation varies with wavelength (optical rotatory dispersion).
- (9) True — Convention.
- (10) True — A racemic acid + a chiral base gives separable diastereomeric salts.
- (11) False — Rapid pyramidal inversion racemises them.
- (12) False — No simple relation exists.
- (13) False — The measured rotation is the same.
- (14) True — Interconversion of anomers.

104. Answer: 9

Read the following 14 statements on special cases and counting of stereoisomers and find how many of them are correct.

Method: judge every statement independently using the definitions of chirality, symmetry elements, configuration and properties of stereoisomers; the explanation given with each statement states why it is true or false.

- (1) True — Axial chirality.
- (2) False — One terminal carbon carries two H atoms.
- (3) True — Ortho-tetrasubstituted biphenyl.
- (4) True — Planar chirality in the strained ring.
- (5) True — Axial chirality of the spiro system.
- (6) False — Three: EE, ZZ and EZ (EZ = ZE).
- (7) True — cis (meso) plus a trans pair.
- (8) False — cis and trans forms are both achiral.
- (9) True — Two unlike stereocentres.
- (10) True — Three unlike stereocentres.
- (11) False — The central carbon carries two identical $\text{-CH}_2\text{OH}$ groups.
- (12) True — Two meso forms and one pair of enantiomers.
- (13) True — $\text{E/Z} \times \text{R/S}$.
- (14) False — It has no stereoisomerism.

Part M — Statements (mixed topics)**105. Answer: 9**

Read the following 14 statements on pairs of stereoisomers, enantiomers and diastereomers and find how many of them are correct.

Method: convert both members to the same representation and compare configuration at every stereocentre: all centres identical $\rightarrow$ same compound (homomers); all centres opposite $\rightarrow$ enantiomers; some same and some opposite $\rightarrow$ diastereomers; different connectivity $\rightarrow$ constitutional isomers. Then apply the property asked (enantiomers share all physical properties except the sign of rotation; diastereomers and constitutional isomers differ).

- (1) True — Configurational isomers (enantiomers, diastereomers) are stereoisomers.
- (2) True — Stereoisomers that are not mirror images.
- (3) False — Constitutional isomers differ in connectivity; stereoisomers do not.
- (4) False — Tautomers are constitutional isomers.
- (5) True — Conformational isomers are stereoisomers.
- (6) True — Non-superimposable mirror images.
- (7) False — Positional isomers are constitutional isomers.
- (8) True — Definition.
- (9) True — Definition.
- (10) True — Having an enantiomer is the definition of chirality.
- (11) False — cis-But-2-ene has a diastereomer (trans) but is achiral.
- (12) True — No internal symmetry element is possible.
- (13) False — No relation between R/S and the sign of rotation.
- (14) True — Not mirror images.

106. Answer: 9

Read the following 14 statements on meso compounds, D/L configuration and sign of rotation and find how many of them are correct.

Method: a meso compound has two or more stereocentres but is achiral because of an internal plane or centre of symmetry. Look for stereocentres arranged symmetrically with opposite configurations.

- (1) True — Internal compensation.
- (2) False — It does have stereocentres.
- (3) True — Achiral.
- (4) True — Internal compensation.
- (5) False — That describes a racemic mixture (external compensation).
- (6) False — D/L is unrelated to the sign of rotation.
- (7) True — E.g. D-(+)-glucose and D-(-)-fructose.
- (8) False — E.g. L-(+)-lactic acid.
- (9) True — Same reasoning.
- (10) True — Fact.
- (11) True — Fact.
- (12) True — (S)-(+)-lactic acid.

(13) True — Mirror-related halves.

(14) False — E.g. meso-2,3,4,5-tetrachlorohexane has four.

107. Answer: 9

Read the following 14 statements on stereogenic atoms other than carbon, spiranes, biaryls and terphenyls and find how many of them are correct.

Method: axial chirality needs a non-planar axis with two different groups at each end (allenes), or ortho substituents large enough to block rotation with two different ortho groups on each ring (biaryls), or two substituted rings on a spiro atom.

- (1) True — S is a stable stereocentre.
- (2) False — Two identical groups on S.
- (3) False — Rapid pyramidal inversion.
- (4) True — No inversion possible.
- (5) True — Resolved by Prelog and Wilen.
- (6) True — High inversion barrier.
- (7) True — Four different groups and no inversion.
- (8) True — Axial chirality.
- (9) True — Restricted rotation about the biaryl bond.
- (10) False — Identical ortho groups on each ring give a plane of symmetry.
- (11) True — Pair of enantiomers + meso.
- (12) True — P and M enantiomers.
- (13) False — Allenes, biaryls and spiranes are chiral without one.
- (14) False — Both termini carry two identical H atoms.

108. Answer: 8

Read the following 14 statements on projection formulas (Fischer, Newman, sawhorse, wedge) and find how many of them are correct.

Method: convert both members to the same representation and compare configuration at every stereocentre: all centres identical → same compound (homomers); all centres opposite → enantiomers; some same and some opposite → diastereomers; different connectivity → constitutional isomers. Then apply the property asked (enantiomers share all physical properties except the sign of rotation; diastereomers and constitutional isomers differ).

- (1) True — Allowed operation.
- (2) True — Not allowed (inverts).
- (3) False — One interchange inverts it.
- (4) True — Even number of swaps.
- (5) False — This inverts the configuration.
- (6) True — Convention.
- (7) True — Convention.
- (8) False — They are conformations.
- (9) True — Different perspectives.
- (10) False — It represents an eclipsed conformation.
- (11) True — Convention.
- (12) True — One swap.
- (13) False — It is independent of the drawing.
- (14) False — Used for acyclic chains.

109. Answer: 10

Read the following 14 statements on resolution, racemic modifications and optical purity and find how many of them are correct.

Method: use $\alpha = [\alpha] \cdot l \cdot c$ (l in dm, c in g/mL), optical purity (OP) = $([\alpha]_{\text{mixture}}/[\alpha]_{\text{pure}}) \times 100$ = enantiomeric excess = %major – %minor, and for mixtures $[\alpha]_{\text{mix}} = \Sigma(\text{mole fraction} \times [\alpha])$.

- (1) True — Three types.
- (2) True — Pasteur's method.
- (3) True — Principle of chemical resolution.
- (4) True — Final step.
- (5) True — Definition.
- (6) False — It is 80%.
- (7) True — Diastereomeric interactions with the phase.
- (8) True — Linear relation.
- (9) False — Equal and opposite contributions cancel.
- (10) False — Equal magnitude, opposite sign.

(11) True — Different compounds.

(12) True — E.g. D-glucose and D-galactose.

(13) True — Chiral active sites.

(14) False — The (+)-acid is still present in excess of its enantiomer.

110. Answer: 10

Read the following 14 statements on cyclic and bridged compounds and find how many of them are correct.

Method: judge every statement independently using the definitions of chirality, symmetry elements, configuration and properties of stereoisomers; the explanation given with each statement states why it is true or false.

- (1) True — Plane of symmetry.
- (2) True — No plane of symmetry.
- (3) False — Plane of symmetry.
- (4) True — No plane of symmetry.
- (5) True — Plane through C2 and C5.
- (6) False — It gives another conformer of the same enantiomer.
- (7) True — Rapid interconversion.
- (8) True — cis (meso) + trans pair.
- (9) True — Plane of symmetry.
- (10) False — They are configurational isomers (diastereomers).
- (11) True — Four stereoisomers.
- (12) True — Planar chirality.
- (13) False — Too strained.
- (14) True — Planes of symmetry.

111. Answer: 10

Read the following 14 statements on conformations and optical activity and find how many of them are correct.

Method: judge every statement independently using the definitions of chirality, symmetry elements, configuration and properties of stereoisomers; the explanation given with each statement states why it is true or false.

- (1) True — The enantiomeric conformers interconvert rapidly.
- (2) True — C_i point symmetry (considering heavy atoms).
- (3) True — Torsional barrier.
- (4) False — Barrier ≈ 14 kJ/mol only.
- (5) True — Conformers average out.
- (6) True — Its gauche rotamers come in enantiomeric pairs.
- (7) False — Configuration is unchanged.
- (8) True — Restricted rotation.
- (9) True — Definition.
- (10) True — Fact.
- (11) False — Anti is more stable.
- (12) True — Fact.
- (13) True — Fact.
- (14) False — Some pairs are 120° apart.

112. Answer: 8

'A pair of stereoisomers might be classified in various ways depending on their exact nature.' Read the following 14 statements and find how many of them could correctly describe a pair of stereoisomers (assuming their nature permits).

Method: judge every statement independently using the definitions of chirality, symmetry elements, configuration and properties of stereoisomers; the explanation given with each statement states why it is true or false.

- (1) True — Two diastereomeric meso forms exist, e.g. in 1,2,3-trichlorocyclohexane.
- (2) False — Tautomers are constitutional isomers.
- (3) True — Definition.
- (4) True — Definition.
- (5) True — Conformers are stereoisomers.
- (6) False — Stereoisomers have the same connectivity.
- (7) True — Enantiomers and diastereomers.
- (8) True — cis-trans pairs.
- (9) True — Enantiomers.
- (10) True — Rotamers are conformers.
- (11) False — A constitutional relationship.
- (12) False — A constitutional relationship.

(13) False — Homomers are identical, not isomers.

(14) False — A constitutional relationship.

Part N — Passage-based questions

113. Answer: 2

PASSAGE. In ortho-substituted biaryls, rotation about the aryl–aryl bond is hindered when each ring carries two different ortho groups, so the bond becomes a stereogenic axis. In the terphenyl (a) shown, the central ring carries four methyl groups and each outer ring carries the ortho groups COOH and CH₃. Question 1: How many stereogenic axes does (a) contain?

Method: axial chirality needs a non-planar axis with two different groups at each end (allenes), or ortho substituents large enough to block rotation with two different ortho groups on each ring (biaryls), or two substituted rings on a spiro atom.

Each aryl–aryl bond has two different ortho groups on both sides: 2 axes.

114. Answer: 3

(Passage, Question 2) How many stereoisomers are possible for the terphenyl (a)?

Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.

Axes can each be (P) or (M): (P,P) and (M,M) are enantiomers; (P,M) = (M,P) is meso: 3.

115. Answer: 2

(Passage, Question 3) How many of the stereoisomers of (a) are optically active?

Method: apply the rules for stereogenic axes: each correctly substituted aryl–aryl bond is one axis; with n independent axes there are 2^n combinations, reduced when the ends are identical (a meso form appears).

(P,P) and (M,M); the (P,M) form has a plane of symmetry.

116. Answer: 1

(Passage, Question 4) How many of the stereoisomers of (a) are meso (achiral)?

Method: a molecule is chiral only when it has no improper symmetry element (plane, centre, S_n). Find the symmetry elements of each structure; if none of POS, COS and AAOS is present the structure is chiral.

Only the (P,M) form.

117. Answer: 1

(Passage, Question 5) How many pairs of enantiomers exist among the stereoisomers of (a)?

Method: convert both members to the same representation and compare configuration at every stereocentre: all centres identical → same compound (homomers); all centres opposite → enantiomers; some same and some opposite → diastereomers; different connectivity → constitutional isomers. Then apply the property asked (enantiomers share all physical properties except the sign of rotation; diastereomers and constitutional isomers differ).

(P,P)/(M,M).

118. Answer: 4

(Passage, Question 6) If one outer ring of (a) is changed to carry the ortho groups NO₂ and CH₃ instead (the other stays COOH/CH₃), how many stereoisomers are possible?

Method: count the stereo-elements (stereogenic centres and double bonds that show E/Z), write 2^n and correct for symmetry (meso forms, identical halves) — equivalent to enumerating the stereoisomers explicitly, as done here by computer enumeration.

Two independent stereogenic axes with unlike ends: $2^2 = 4$ (two enantiomeric pairs).